

Shariah Computational Financial Engineering I — Reconstructing Retail FX and CFD Market Access: A Structural Incompatibility Analysis and First-Principles Alternative Architecture

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Abstract

Retail contracts for difference (CFDs) and leveraged foreign exchange (FX) trading constitute one of the fastest-growing segments of global retail market access, with daily notional turnover exceeding several trillion United States dollars across major brokerages. Despite the size of the observant-Muslim retail population, no rigorous mathematical treatment of the compatibility between the standard CFD/FX architecture and Islamic contract law currently exists in the peer-reviewed literature. Drawing on primary-source observations from a Capital Markets and Trading Analyst residency at Bacera Co Pty Ltd, this paper formalises the CFD payoff mechanism, the Tom-Next overnight financing structure, and the leverage–margin dynamics as a coupled stochastic system driven by a Bates (1996) jump-diffusion with stochastic volatility. Within this formal model we document five structural incompatibilities or concerns between the standard retail CFD and the selected AAOIFI/OIC constraint framework: an interest-rate-linked financing mechanism (*ribā* concern), a synthetic contingent payoff without ownership or delivery (*gharar* concern), a counterparty-reciprocity structure under B-book execution (*maysir* concern), non-fit with the classical nominate contract categories including

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absence of *qabd*, and a reciprocal-deferred-contingent obligation structure (*bay' al-kāli' bi'l-kāli'* concern). We further show that the “swap-free Islamic account” variant marketed by conventional brokers addresses at most the first of these concerns and only in part. We then construct, subject to the same constraint framework, three candidate alternative market-access architectures: (i) a *wa'd*-based FX exposure vehicle, whose scholarly permissibility depends on whether the double-*wa'd* structure is admitted under the recognised authorities (this paper takes the AAOIFI restrictive reading and records the divergence); (ii) a commodity-*murābaḥa* financing overlay, whose organised form is impermissible under OIC Resolution 179; and (iii) a *sharīkat al-'inān* pooled-equity structure that is formally consistent with the operational design constraints adopted in this paper, subject in every case to review by a qualified Shariah supervisory board. For each we derive the payoff dynamics and holding cost under the same underlying process. A simulation-based quantitative comparison, under illustrative Bates parameterisations consistent with published FX and equity implied-volatility studies, quantifies the fee-structure differential, tail-risk properties, and margin-call trigger distributions of each architecture against a conventional leveraged CFD baseline. Because the *sharīkat* architecture cannot be leveraged within the joint-ownership construction adopted, the comparison is properly understood as a quantification of the economic cost of the leverage cap imposed by that construction: the compliant architecture forgoes proportional leveraged upside and leveraged downside relative to a CFD of matched client capital, while removing the structural concerns identified. The paper does not issue fiqh rulings; it formalises the mathematical structure and documents its consistency, or otherwise, with the adopted constraint framework. Ultimate compliance certification remains the province of qualified Shariah supervisory boards. An open-source Python implementation is released with the paper, together with a full empirical calibration pipeline outlined as future work.

Keywords: Islamic finance, contracts for difference, foreign exchange, stochastic modelling, Bates model, AAOIFI, *sharīkat al-'inān*, *wa'd*, market microstructure, financial engineering

JEL: G13, G14, G23, Z12

1. Introduction

1.1. The retail leverage market and its participants

The retail contracts-for-difference (CFD) and leveraged foreign exchange (FX) segment has grown from an institutional niche in the early 2000s into one of the largest retail-facing derivative markets in the world. The Bank for International Settlements’ 2025 Triennial Central Bank Survey reports global daily over-the-counter FX turnover of approximately USD 9.6 trillion in April 2025 (Bank for International Settlements, 2025); retail leveraged CFD activity is a much smaller subset of this aggregate but is a segment growing more rapidly than the OTC total. Regulators in the European Union, United Kingdom, and Australia have documented that a persistently large fraction, between roughly 74 and 89 percent, depending on jurisdiction and reporting period, of retail CFD accounts lose money over a given year (European Securities and Markets Authority, 2018; Australian Securities and Investments Commission, 2020; Financial Conduct Authority, 2019). Despite these outcomes, retail CFD participation continues to expand, driven by low account minimums, smartphone-native execution platforms, and aggressive marketing.

A material and growing subset of this retail participant base is observant Muslim. There are approximately 2.0 billion Muslims globally, of whom a rising share are working-age and financially active (Pew Research Center, 2015). Refinitiv’s *Islamic Finance Development Report 2023* estimates global Islamic finance assets at approximately USD 4.5 trillion, projected to reach USD 6.7 trillion by 2027 (Refinitiv and Islamic Corporation for the Development of the Private Sector, 2023). Yet the overwhelming majority of these assets sit in institutional vehicles, Islamic banking, *ṣukūk*, *takāful*, and Shariah-compliant real estate, rather than in retail market-access products for short-horizon leveraged synthetic exposure of the kind associated with CFDs. Muslim retail participants seeking such exposure currently face limited options: participate in conventional CFD/FX markets under some implicit compromise of religious observance, use a “swap-free Islamic account” variant marketed by conventional brokers (which is examined and shown structurally insufficient in Section 4), or forgo leveraged synthetic exposure and confine activity to permissible spot vehicles. Alternatives such as Shariah-compliant equity funds, spot equity investment under AAOIFI screening, permissible spot commodity transactions, and *ṣukūk* are available for other investment purposes but do not substitute for the specific short-horizon leveraged synthetic

exposure use case.

1.2. The scholarly gap

The academic literature on Islamic finance is extensive on the topics of *ṣukūk* pricing (Çakır and Raei, 2007; Jobst, 2007), *murābaḥa* contract structure (Usmani, 2002), Islamic equity screening (Derigs and Marzban, 2008; Kunhibava and Shanmugam, 2010), and *takāful* risk-sharing (Iqbal and Mirakhor, 2011). It is comparatively silent on the intersection of retail derivatives, market microstructure, and Islamic contract law. The closest prior work addresses institutional Islamic derivatives (DeLorenzo, 2007; Bacha, 1999) and Islamic hedging frameworks based on *wa’d* promises (Bacha and Mirakhor, 2013), but does not formally treat the retail CFD architecture, the Tom-Next financing mechanism, or the specific structural violations they generate.

A structured literature search was conducted across Scopus, Web of Science, Google Scholar, SSRN, and the Islamic Financial Services Board and ISRA repositories, using combinations of the terms {“Contract for Difference”, “CFD”, “leveraged retail FX”, “Tom-Next”, “rollover swap”} with {“Shariah”, “Islamic finance”, “AAOIFI”, “*ribā*”, “*gharar*”, “*maysir*”, “*qabḍ*”}, covering publications indexed through January 2026. No study matching all three of the following criteria was identified: (i) formalising the retail CFD contract as a fiqh-constrained stochastic system with explicit price-process, financing, and margin dynamics; (ii) proving the resulting structural concerns from primary fiqh sources at the level of formal propositions; and (iii) constructing and quantitatively comparing first-principles alternative architectures under the constraint set implied by AAOIFI Shariah Standards. This paper addresses that gap. The novelty claim is narrowed accordingly: it is an assertion about the peer-reviewed literature indexed in the named databases through the stated date, not a universal claim about all extant scholarship.

1.3. Research questions

The paper is organised around three research questions:

- RQ1.** What structural conditions must a leveraged retail market-access contract satisfy to be candidates for review as consistent with the requirements of AAOIFI Shariah Standards No. 1 (Trading in Currencies), No. 8 (*Murābaḥa*), No. 12 (*Sharīkah*), and No. 18 (*Qabḍ*)?

- RQ2.** Under what conditions, if any, can these constraints be simultaneously satisfied for the economic use cases currently served by retail CFDs, namely short-horizon directional exposure, hedging of underlying commercial risk, and cash-efficient synthetic exposure?
- RQ3.** What is the economic cost differential between a Shariah-compliant alternative architecture and a conventional leveraged CFD, and how does this differential vary with holding horizon, underlying volatility, and leverage regime?

1.4. Contributions

The paper makes five contributions.

1. A formal mathematical characterisation of the standard CFD/FX architecture as a stochastic contract system, including the price-process, Tom-Next financing rate, leverage–margin dynamics, and broker revenue decomposition (Section 3).
2. Formal propositions documenting five structural incompatibilities or concerns between the standard CFD and the adopted AAOIFI/OIC constraint framework, together with a proof that swap-free Islamic accounts address at most one of these and only in part (Section 4).
3. Three first-principles alternative architectures, *wa'd*-based FX exposure, commodity-*murābaḥa* financing overlay, and *sharīkat al-ʿinān* pooled-equity structure, with explicitly derived payoff dynamics under a Bates jump-diffusion price process (Section 5).
4. A Monte Carlo comparison of cost-of-carry, tail-risk, and margin-call trigger distributions across architectures under illustrative Bates parameterisations, honestly framed as a quantification of the compliance-cost of the Shariah-imposed leverage cap rather than a claim of intrinsic economic superiority of one architecture over another (Section 6).
5. An open-source Python reference implementation, released with the paper under a permissive licence, enabling replication and extension.

1.5. Positioning and limitations

This paper is a work of mathematical financial engineering informed by primary fiqh sources. It is not a work of fiqh. The author is a mathematics and statistics student, not a qualified Shariah scholar. Every claim of structural incompatibility in Section 4 is derived by citation to a specific AAOIFI Shariah

Standard, a resolution of the International Islamic Fiqh Academy of the Organisation of Islamic Cooperation (OIC), or a widely accepted classical or contemporary fiqh source, and is stated as a formal proposition subject to that citation. The alternative architectures proposed in Section 5 are candidate structures whose ultimate certification for practical use remains the province of qualified Shariah supervisory boards. Where scholarly opinion differs between the recognised schools, the paper states the divergence explicitly and does not attempt to resolve it.

The paper is further limited to the retail CFD/FX segment. Institutional Islamic derivatives, *ṣukūk*, *takāful*, and Islamic banking are treated only insofar as they inform the market-access architectures under consideration. The Monte Carlo analysis in Section 6 operates under illustrative Bates parameterisations rather than a live empirical calibration; a full calibration pipeline against real tick data is outlined in Appendix Appendix D and deferred to a subsequent paper in this series. Extension to order-book microstructure, adverse selection, and regime-switching dynamics is similarly deferred.

1.6. Structure of the paper

Section 2 surveys the relevant Islamic finance and market-microstructure literatures and identifies the specific gap this paper addresses. Section 3 formalises the standard CFD/FX architecture and its stochastic mechanics. Section 4 states and proves the five structural incompatibilities or concerns with the adopted AAOIFI/OIC constraint framework. Section 5 constructs three candidate alternative architectures from first principles. Section 6 presents the Monte Carlo comparison. Section 7 discusses limitations, regulatory and implementation considerations, and open questions. Section 8 concludes and identifies directions for subsequent papers in this series.

2. Literature Review

This section surveys the three literatures on which the paper draws: (i) Islamic commercial law and its foundational nominate contracts, (ii) the extant scholarly and academic work on Islamic derivatives and hedging, and (iii) mainstream retail market-microstructure and trader-behaviour research. The three streams are treated in turn and the specific gap the paper addresses is stated formally in Section 2.4.

2.1. Islamic commercial law and the foundational contracts

Islamic commercial law (*fiqh al-mu'āmalāt*) is derived from the four canonical sources of the Shariah, the Qur'ān, the Sunnah of the Prophet Muḥammad, *ijmā'* (juristic consensus), and *qiyās* (analogical reasoning), and codified across the received bodies of the four Sunnī schools (Ḥanafī, Mālikī, Shāfi'ī, Ḥanbalī) and the Ja'farī school (Kamali, 2000; Vogel and Hayes, 1998). Contemporary reference works synthesise this corpus into a manageable set of nominate contracts that recur across modern Islamic finance practice (Usmani, 2002; El-Gamal, 2006; Ayub, 2007; Iqbal and Mirakhor, 2011).

The nominate contracts most relevant to the market-access architectures considered in this paper are the following.

- *Bay'* (sale), the exchange of one item of property for another with mutual consent; a valid *bay'* requires *māl* (property of recognised value), *qabḍ* (constructive or physical possession by the seller prior to sale; see AAOIFI Shariah Standard No. 18), and clarity as to subject-matter, quantity and price (Accounting and Auditing Organization for Islamic Financial Institutions, 2015c).
- *Salam*: a forward sale in which the price is paid in full at contract inception against deferred delivery of a fungible commodity of specified quality and quantity.
- *Istiṣnā'*, a commission-to-manufacture contract for a specified good to be produced or built to specification.
- *Murābaḥa*, a cost-plus disclosed-markup sale, widely deployed as the workhorse financing structure of Islamic banking (AAOIFI Standard No. 8; Usmani, 2002).
- *Mushāraka*, a joint-venture partnership with proportional profit-and-loss sharing (AAOIFI Standard No. 12).
- *Sharīkat al-'inān*, a subclass of *mushāraka* in which each partner contributes capital and, optionally, labour; profit sharing is by pre-agreed ratio, but loss sharing must be strictly proportional to capital contribution.
- *Mudāraba*, a silent-partnership contract in which one party (*rabb al-māl*) provides capital and the other (*mudārib*) provides labour and management.

- *Ijāra*, a lease of the usufruct of an asset with ownership risk retained by the lessor.
- *Wakāla*, an agency contract, in which one party acts for another for a defined fee (*ujra*).
- *Wa'd*: a unilateral promise; classically treated as a moral rather than legal obligation, but binding on the promisor under contemporary AAOIFI and Bank Negara Malaysia rulings when structured as a formal, one-sided undertaking.

Every valid contract in this framework must simultaneously exclude three prohibited features: *ribā* (usury and, by extension, contractual interest), *gharar* (excessive uncertainty as to subject-matter, price, or execution), and *maysir* (gambling, understood as a zero-sum transaction in which wealth transfers by pure chance). The prohibitions are grounded in explicit Qur'ānic verses, 2:275 for *ribā*, 5:90 for *maysir*, and in Prophetic ḥadīth for *gharar*, and are treated as absolute constraints rather than as variables in a trade-off (Chapra, 1985; Warde, 2010). Each prohibition admits formal characterisation as a constraint on the parameters of a contract; this observation is exploited in Section 4 to state structural violations of the standard CFD as formal propositions.

The consolidation of contemporary practice around a standardised set of contract types has been driven principally by the Accounting and Auditing Organization for Islamic Financial Institutions (AAOIFI), whose *Shari'ah Standards* (Accounting and Auditing Organization for Islamic Financial Institutions, 2015c) are adopted or referenced in the regulatory frameworks of Bahrain, the United Arab Emirates, Qatar, Oman, Jordan, Sudan, Syria, Nigeria, and, in modified form, Malaysia and Indonesia; and by the resolutions of the International Islamic Fiqh Academy of the Organisation of Islamic Cooperation (International Islamic Fiqh Academy, 1992, 2009). This paper adopts AAOIFI Standards as its reference framework, on the ground that they represent the most widely subscribed international codification of Islamic commercial law and are cross-referenced in nearly every peer-reviewed article on the subject.

2.2. The Islamic derivatives and hedging debate

The Shariah status of derivative instruments is one of the most actively contested subjects in contemporary Islamic finance scholarship. The debate

divides broadly along two axes: (i) whether risk-transfer instruments as a class are compatible with the Islamic requirement that commercial gain be linked to real economic activity, and (ii) whether specific instruments, futures, options, swaps, contracts for difference, can be restructured so as to satisfy formal fiqh constraints.

The *restrictive* position, associated with Usmani (2002) and with the majority of scholars in the Gulf Cooperation Council (GCC), holds that conventional derivatives violate *gharar* through the uncertainty of settlement embedded in their cash-settled payoff, violate *maysir* through their zero-sum counterparty structure, and, in the case of leveraged products, violate the *ribā* prohibition through their financing overlay. On this view, most conventional derivatives cannot be Islamised through structural modification: only fundamentally different instruments (*salam*, *istiṣnāʾ*, and equity-based partnership structures) satisfy the constraints. Kamali (2000) provides the most sustained scholarly defence of the restrictive position with respect to futures and options specifically.

The *permissive* position, associated principally with Bank Negara Malaysia's Shariah Advisory Council and with a subset of scholars in Malaysia, Indonesia, and the ISRA (International Shari'ah Research Academy) tradition, holds that certain derivative economic functions, hedging of underlying commercial exposure, transfer of specific risks, may be permissibly discharged through *wa'd*-based structures and combinations of nominate contracts (Bacha and Mirakhor, 2013; Dusuki, 2011; International Shari'ah Research Academy for Islamic Finance, 2016). The Islamic Profit Rate Swap and the Islamic FX Forward, both developed and deployed by Malaysian and Gulf Islamic banks, are the two most prominent instruments to have emerged from this school.

AAOIFI sits closer to the restrictive pole. Its Shariah Standard No. 1 explicitly prohibits currency forwards and options; Standard No. 20 restricts organised commodity markets; and Standard No. 27 governs indices. The International Islamic Fiqh Academy of the OIC has issued Resolution No. 63 (1/7) ruling against conventional forwards and options, and Resolution No. 179 (19/5) ruling against organised *tawarruq* (International Islamic Fiqh Academy, 1992, 2009). Where a candidate architecture proposed in this paper is more permissive on any dimension than AAOIFI, the divergence is stated explicitly and the affected result is qualified accordingly.

Prior academic treatments of specific instrument classes include: De-Lorenzo (2007) on the total-return swap and its use of *wa'd*-based "Shariah conversion technology"; Bacha (1999) on the general permissibility archi-

tructure of derivatives; Jobst (2007) on Islamic securitisation, with a formal treatment of *ṣukūk* structuring under a real-asset requirement; Kunhibava and Shanmugam (2010), whose comparative analysis of Shariah and conventional-law objections to derivatives is the closest general survey in the peer-reviewed literature; Al-Suwailem (2006) on hedging under Islamic constraints; and Uberoi and Evans (2008) on the practitioner mechanics of *wa'd*-based hedging. On the retail-facing use case of leveraged FX and CFDs specifically, however, the peer-reviewed literature is essentially silent, this is the gap identified formally in Section 2.4.

2.3. Retail market microstructure and trader behaviour

A rigorous analysis of retail CFD/FX market access requires grounding in the mainstream financial-economics literature on market microstructure, retail trader behaviour, and leverage regulation. This grounding is necessary for two reasons. First, it anchors the paper's conclusions on cost-of-carry, execution asymmetry, and margin-call dynamics in an established empirical evidence base rather than in fiqh reasoning alone. Second, it locates the paper within the mainstream finance conversation, so that its findings are legible, and falsifiable, to reviewers and readers outside the Islamic finance subfield.

The foundational microstructure literature on informed and uninformed order flow, adverse selection, and dealer market-making descends from Kyle (1985) and Glosten and Milgrom (1985), with the standard practitioner treatment given in Harris (2003) and the theoretical consolidation in O'Hara (1995). The retail-broker-specific extension of this framework, developed over the last decade around the economics of "payment for order flow" and the "B-book" business model, is surveyed in Menkveld (2016). Together these works provide the theoretical basis for the broker-revenue decomposition introduced in Section 3, in which a fraction $\beta \in [0, 1]$ of client order flow is internalised by the broker (B-book execution) rather than externalised to an exchange or liquidity provider (A-book execution), with the implication that broker profit is, at least in part, the algebraic mirror of client loss.

The retail-trader behaviour literature is even sharper. Barber and Odean (2000, 2013) established, using account-level data on tens of thousands of retail brokerage clients, that individual investors as a class systematically underperform passive benchmarks net of transaction costs. Grinblatt and Keloharju (2000) confirm the pattern for the Finnish retail population. Barber et al. (2014), using account-level data on the near-universe of Taiwanese day traders, find that fewer than one per cent of day traders earn returns

sufficient to cover transaction costs over any sustained period, and that the top and bottom performers exhibit substantial persistence. Kelley and Tetlock (2013) find that retail order flow is predominantly liquidity-motivated and, on aggregate, adversely selected. The consistency of these findings across markets and decades is difficult to overstate.

Leverage-specific regulation has generated its own body of evidence. Heimer and Simsek (2019) evaluate the 2010 U.S. Commodity Futures Trading Commission limit on retail FX leverage, from 100:1 to 50:1, and find that the intervention meaningfully reduced retail losses without materially reducing participation. The European Securities and Markets Authority intervention against retail CFDs (European Securities and Markets Authority, 2018), the Financial Conduct Authority policy statement PS19/18 (Financial Conduct Authority, 2019), and the Australian Securities and Investments Commission Product Intervention Order (Australian Securities and Investments Commission, 2020) each report retail CFD account loss rates in the 74–89 per cent range and impose leverage caps, margin-close-out requirements, and negative-balance protection accordingly. These findings are treated in this paper as external constraints on the design space for any candidate retail market-access architecture, Shariah-compliant or otherwise: a compliant architecture that leaves retail participants materially worse off in expectation than the conventional benchmark is not a socially defensible design, whatever its jurisprudential merits.

2.4. The identified gap

The three literatures surveyed above have not, to the best of the author’s knowledge, been synthesised in the peer-reviewed record. The Islamic commercial-law and derivatives literatures address the contractual and jurisprudential dimensions but do not treat the retail CFD/FX architecture as a formal stochastic system, do not derive its specific structural violations at the level of formal propositions, and do not construct alternative architectures whose payoff and cost-of-carry dynamics are explicitly modelled under a common underlying price process. The market-microstructure and retail-behaviour literatures treat the empirical mechanics of retail CFD and FX markets rigorously but make no reference to Islamic contract law, or to the substantial observant-Muslim retail participant base that intersects those markets. Practitioner treatments of “Islamic FX” and “swap-free accounts”, where they exist in the trade press and in broker marketing material, are unsourced, and are shown in Section 4 to be structurally insufficient.

The specific gap this paper addresses is therefore stated as follows: *no prior peer-reviewed work treats the retail CFD/FX architecture as a formal fiqh-constrained stochastic system, proves its structural violations at the level of formal propositions grounded in AAOIFI Shariah Standards and OIC Fiqh Academy resolutions, or constructs and quantitatively compares first-principles alternative architectures under the resulting constraint set.* The remainder of the paper addresses this gap directly: Section 3 formalises the CFD/FX architecture; Section 4 documents its five structural violations; Section 5 constructs three candidate alternative architectures; and Section 6 compares them quantitatively.

3. Formal Characterisation of the CFD/FX Architecture

This section formalises the standard retail CFD/FX architecture as a coupled stochastic system, comprising: an underlying price process (Section 3.1), a contract-payoff mapping (Section 3.2), a financing mechanism (Section 3.3), a margin and leverage dynamic (Section 3.4), and a broker-revenue decomposition (Section 3.5). The section concludes with primary-source institutional context from a four-week Capital Markets and Trading Analyst residency at Bacera Co Pty Ltd (Section 3.6). The characterisation established here is the object of the Shariah analysis in Section 4.

3.1. Preliminaries and the underlying price process

Throughout the paper we work on a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \geq 0}, \mathbb{P})$ satisfying the usual conditions of right-continuity and $\mathbb{P}$ -completeness. All stochastic processes are adapted to $\{\mathcal{F}_t\}_{t \geq 0}$. The measure $\mathbb{P}$ denotes the physical (real-world) probability measure; risk-neutral counterparts, where required, are denoted $\mathbb{P}^*$ or $\mathbb{Q}$.

Let $(S_t)_{t \geq 0}$ denote the spot price of the underlying asset at time t . For an FX pair CCY1/CCY2 (*e.g.* EUR/USD), S_t denotes the price of one unit of the base currency CCY1 quoted in units of the quote currency CCY2. For an equity or index underlying, S_t denotes the spot price in the reference currency.

Assumption 3.1 (Underlying price process). The underlying price process $(S_t)_{t \geq 0}$ follows the Bates (1996) jump-diffusion model with stochastic volatility.

The dynamics under $\mathbb{P}$ are given by the coupled system

$$\frac{dS_t}{S_{t-}} = (\mu - \lambda\kappa) dt + \sqrt{v_t} dW_t^S + (J_t - 1) dN_t, \quad (1)$$

$$dv_t = \kappa_v(\theta - v_t) dt + \sigma_v \sqrt{v_t} dW_t^v, \quad (2)$$

where:

- $\mu \in \mathbb{R}$ is the drift under $\mathbb{P}$; under an equivalent martingale measure $\mathbb{P}^*$ the drift is replaced by $r - q$, with r the domestic risk-free rate and q the dividend yield (for equity) or foreign risk-free rate (for FX);
- $(v_t)_{t \geq 0}$ is the instantaneous variance, following the Heston (1993) square-root process with mean-reversion speed $\kappa_v > 0$, long-run variance $\theta > 0$, and volatility-of-volatility $\sigma_v > 0$;
- W^S and W^v are standard Brownian motions with correlation $d\langle W^S, W^v \rangle_t = \rho dt$, $\rho \in [-1, 1]$;
- N_t is a Poisson process with intensity $\lambda > 0$, independent of W^S and W^v ;
- $(J_t)_{t \geq 0}$ is a sequence of i.i.d. strictly positive relative jump sizes, with $\ln J_t \sim \mathcal{N}(\mu_J, \sigma_J^2)$;
- $\kappa := \mathbb{E}[J_t - 1] = \exp(\mu_J + \frac{1}{2}\sigma_J^2) - 1$ is the mean relative jump size, subtracted from the drift so that the compensated jump term is a $\mathbb{P}$ -martingale.

The strict Feller condition $2\kappa_v\theta > \sigma_v^2$ is imposed throughout to guarantee that the variance process v_t almost surely does not attain zero (Heston, 1993); the boundary case $2\kappa_v\theta = \sigma_v^2$ is excluded.

Remark 3.1 (Measure choice for the Shariah and simulation analysis). The Shariah structural analysis (Section 4) depends only on the pathwise almost-sure properties of (S_t) , which are invariant under equivalent measure change and therefore hold under both $\mathbb{P}$ and any equivalent $\mathbb{P}^*$. The Monte Carlo simulations (Section 6) are conducted under the physical measure $\mathbb{P}$ throughout, using the parameters of Table 3, to reproduce empirically observed loss distributions, margin-call frequencies and Sharpe ratios of the

sort a retail client would face. Where a risk-neutral valuation is required (Proposition 5.1), we invoke the standard equivalent martingale measure for the incomplete jump-diffusion market, noting that under $\mathbb{P}^*$ the jump intensity λ and the jump-size law of $\ln J_t$ may differ from their $\mathbb{P}$ -counterparts; the choice of $\mathbb{P}^*$ within the family of equivalent measures is not unique in an incomplete market (Cont and Tankov, 2004). No claim in the paper turns on a specific such choice.

Remark 3.2 (Choice of price process). The Bates model is selected over the plain Black and Scholes (1973) geometric Brownian motion for three reasons load-bearing for the analysis of Sections 4–6. First, retail CFD/FX markets display well-documented jump behaviour around scheduled macroeconomic releases (US non-farm payrolls, FOMC and ECB decisions, RBA cash-rate announcements) and around unscheduled events (central bank interventions, flash crashes such as the CHF un-pegging of 15 January 2015); these jumps drive the clustering of margin-call trigger events, which the diffusion-only benchmark cannot reproduce (Merton, 1976). Second, the leverage effect, empirically, $\rho < 0$ for equity underlyings, is quantitatively material for tail-risk assessment and is captured by the correlated Brownian structure. Third, stochastic volatility is required to reproduce the implied volatility smiles observed in traded FX and equity option markets against which the model would be calibrated (Heston, 1993; Björk, 2009).

3.2. The contract for difference and its payoff process

We now formalise the CFD contract as a mapping from the underlying price path to a cash-settled payoff to the client, together with a financing overlay to be specified in Section 3.3.

Definition 3.1 (Contract for Difference). A *contract for difference* (CFD) on the underlying S is a bilateral cash-settled derivative contract between a client (retail participant) and a broker (counterparty), specified by the tuple $(t_0, t_1, \delta, N, L, s, m)$ where:

- $t_0 \in [0, T]$ is the entry time,
- $t_1 \in (t_0, T]$ is the exit time (chosen by the client or triggered exogenously by margin-call; see Section 3.4),
- $\delta \in \{+1, -1\}$ is the directional indicator (+1 for a long position, -1 for a short position),

- $N \in \mathbb{R}_+$ is the notional size, measured in units of the underlying,
- $L \geq 1$ is the leverage ratio,
- $(s_t)_{t \geq 0}$ is the bid–ask spread process, and
- $(m_t)_{t \geq 0}$ is the broker’s markup process over the interbank financing rate (see Section 3.3).

At exit time t_1 the broker settles cash to the client in the amount of the net payoff $\Pi_{\text{net}}(t_0, t_1)$ defined in equation (4) below.

Definition 3.2 (Gross and net payoff). The client’s *gross payoff* on a CFD held from t_0 to t_1 is

$$\Pi_{\text{gross}}(t_0, t_1) = \delta \cdot N \cdot (S_{t_1} - S_{t_0}). \quad (3)$$

The corresponding *net payoff*, accounting for accumulated financing and transaction costs, is

$$\Pi_{\text{net}}(t_0, t_1) = \delta \cdot N \cdot (S_{t_1} - S_{t_0}) - \int_{t_0}^{t_1} F_s \, ds - C(t_0, t_1), \quad (4)$$

where $(F_s)_{s \geq 0}$ is the instantaneous financing rate derived in Section 3.3, and $C(t_0, t_1)$ is the aggregate transaction cost from spreads and fixed commissions,

$$C(t_0, t_1) = N \cdot \frac{s_{t_0} + s_{t_1}}{2} + c_{\text{fixed}}, \quad (5)$$

with $c_{\text{fixed}} \geq 0$ a broker-specific per-trade commission (often zero on retail FX; positive on single-name equity CFDs).

Remark 3.3 (Cash settlement and the absence of delivery). Critical to the fiqh analysis of Section 4: at no point during $[t_0, t_1]$ does the client acquire ownership of, or any legal or beneficial claim to, N units of the underlying S . The client’s exposure is exclusively to the price difference $S_{t_1} - S_{t_0}$. This distinguishes the CFD from (i) a spot purchase of the underlying, which confers ownership; (ii) a forward or futures contract requiring physical delivery; and (iii) an equity share carrying voting and residual-claim rights. The absence of any transfer of ownership is the empirical foundation of the *qabd* violation proved as Proposition 4.4 in Section 4.

3.3. The Tom-Next swap: formal derivation of the financing rate

The overnight rollover fee, industry-termed the “Tom-Next swap” (from “tomorrow-next-day”), is the mechanism by which brokers charge (or credit) the client for holding a leveraged position across the daily settlement cutoff, conventionally 5:00 PM New York time. This is the empirical mechanism underlying the *ribā* violation formalised in Section 4. We derive it here from first principles under covered interest rate parity.

3.3.1. Derivation for FX pairs

Consider a long CFD position of notional N on the FX pair CCY1/CCY2, held from settlement date T_d to the next business settlement date T_{d+1} . Under covered interest rate parity (Björk, 2009, Ch. 24), the no-arbitrage forward exchange rate satisfies

$$F(T_d, T_{d+1}) = S_{T_d} \cdot \frac{1 + r_{T_d}^{\text{CCY2}} \tau}{1 + r_{T_d}^{\text{CCY1}} \tau}, \quad (6)$$

where $r_{T_d}^{\text{CCY}}$ is the overnight interbank lending rate for currency CCY at T_d , SOFR for USD, ESTR for EUR, SONIA for GBP, TONA for JPY, and the RBA cash rate for AUD, and $\tau = 1/365$ is the actual/365 day-count fraction.

Rolling a spot position forward by one day requires the counterparty to pay (or receive) the difference $F(T_d, T_{d+1}) - S_{T_d}$. For small τ , first-order Taylor expansion of (6) gives

$$F(T_d, T_{d+1}) - S_{T_d} = S_{T_d} \cdot (r_{T_d}^{\text{CCY2}} - r_{T_d}^{\text{CCY1}}) \tau + O(\tau^2). \quad (7)$$

The broker charges the client this differential plus a proprietary markup $m_{T_d} \geq 0$. The Tom-Next swap fee for a long CFD position held across the cutoff is therefore

$$F_{T_d}^{\text{swap}} = N \cdot S_{T_d} \cdot \frac{(r_{T_d}^{\text{CCY2}} - r_{T_d}^{\text{CCY1}}) + m_{T_d}}{365}. \quad (8)$$

For a short position ($\delta = -1$), the sign of the interbank differential reverses (since the client is now the funder of the base currency), but the broker’s markup is charged with a sign that always disadvantages the client. Empirically, retail brokers post separate long and short swap rates in which the markup component is applied so that both sides of the market bear a non-negative broker margin (see, e.g., MetaTrader 5 symbol specifications,

which display distinct `swap_long` and `swap_short` figures with the broker markup embedded in each). We encode this sign convention explicitly in the continuous-time definition below by separating the CIP-implied differential (whose direction depends on δ) from the markup (which is always a cost to the client).

Definition 3.3 (Instantaneous financing rate). Passing to the continuous-time limit, the instantaneous financing rate F_s that appears in the net payoff (4) is

$$F_s = N \cdot S_s \cdot \left[\delta (r_s^{\text{CCY2}} - r_s^{\text{CCY1}}) + m_s \right], \quad (9)$$

where r_s^{CCY1} and r_s^{CCY2} are the instantaneous overnight interbank rates for the base and quote currencies respectively, $m_s \geq 0$ is the broker's continuous-time markup process, and $\delta \in \{+1, -1\}$ is the position direction. Under this sign convention, both the long-side and short-side financing charges include a non-negative markup contribution, consistent with observed retail broker pricing.

Proposition 3.1 (Interbank-rate measurability of the financing rate). *Under Definition 3.3, the Tom-Next financing rate F_s is measurable with respect to the σ -algebra $\mathcal{G}_s := \sigma(S_s, r_s^{\text{CCY1}}, r_s^{\text{CCY2}}, m_s)$ generated by the underlying price and the two interbank overnight rates. In particular, conditional on $\{S_s, m_s, \delta\}$, F_s is an affine function of the interbank rate differential $r_s^{\text{CCY2}} - r_s^{\text{CCY1}}$ with sensitivity:*

$$\frac{\partial F_s}{\partial (r_s^{\text{CCY2}} - r_s^{\text{CCY1}})} = \delta \cdot N \cdot S_s. \quad (10)$$

Proof. Direct from equation (9). The coefficients δ , N , S_s , m_s are $\mathcal{F}_s$ -measurable by construction. The rates r_s^{CCY1} and r_s^{CCY2} are observable at time s (published by the respective central banks or interbank rate administrators). The affine-with-unit-sensitivity property is immediate from differentiation of (9) with respect to the differential. $\square$ $\square$

Remark 3.4 (The *ribā* structural implication, foreshadowed). Proposition 3.1 formalises what is empirically obvious to any practitioner: the CFD financing rate is, by construction, a function of interbank interest rates. Under the general *ribā* prohibition articulated in Usmani (2002) and Kamali (2000), namely the prohibition of a fixed increment charged for the deferral of exchange, and taken together with AAOIFI Shariah Standard No. 1 (on Trading

in Currencies) and No. 8 (on *Murābaḥa*), which respectively restrict deferred currency transactions and the addition of stipulated increments to credit-sale contracts, the Tom-Next financing structure is at minimum a serious *ribā* concern under the majority contemporary reading; the argument is developed formally in Section 4 and its full authority basis is discussed there.

3.4. Margin, leverage, and margin-call dynamics

Leverage is central to the retail CFD product: it is the feature that converts small-notional retail capital into large-notional market exposure, and it is the feature that drives the empirically observed loss rates cited in Section 1.1. We formalise its mechanics.

3.4.1. Initial and maintenance margin

For a CFD position of notional value $N \cdot S_{t_0}$ opened at leverage L , the client posts *initial margin* equal to

$$M_0 = \frac{N \cdot S_{t_0}}{L}, \quad (11)$$

which is the fraction $1/L$ of the notional value of the exposure. The broker requires a *maintenance margin* threshold

$$M_{\text{maint}} = \alpha \cdot M_0, \quad \alpha \in (0, 1], \quad (12)$$

with α typically in $[0.25, 0.75]$ depending on regulatory regime and asset class. The client's account equity at time t is

$$E_t = M_0 + \delta \cdot N \cdot (S_t - S_{t_0}) - \int_{t_0}^t F_s \, ds. \quad (13)$$

3.4.2. The margin-call condition

The margin call is triggered at the first stopping time τ_{MC} such that account equity falls at or below the maintenance threshold:

$$\tau_{\text{MC}} := \inf\{t > t_0 : E_t \leq M_{\text{maint}}\}. \quad (14)$$

On short holding horizons the financing term $\int_{t_0}^t F_s \, ds$ is small relative to price innovations and may be neglected in a leading-order analysis. Under

this approximation, the barrier condition (14) simplifies to

$$\delta \cdot N \cdot (S_t - S_{t_0}) \leq -(1 - \alpha)M_0, \quad (15)$$

which for a long position ($\delta = +1$), combined with (11), yields the explicit price barrier

$$S_t \leq S_{t_0} \cdot \left(1 - \frac{1 - \alpha}{L}\right). \quad (16)$$

For a short position the barrier lies symmetrically above S_{t_0} .

Proposition 3.2 (Leverage sensitivity of margin-call probability). *Suppose S_t follows a geometric Brownian motion with constant volatility σ (Assumption 3.1 with $\lambda = 0$ and $v_t \equiv \sigma^2$). For a long CFD position with leverage L and holding horizon $h := t_1 - t_0$, the margin-call probability admits the approximation*

$$\mathbb{P}(\tau_{MC} < t_1 \mid \mathcal{F}_{t_0}) \approx 2\Phi\left(-\frac{1 - \alpha}{L\sigma\sqrt{h}}\right), \quad (17)$$

where Φ denotes the standard normal cumulative distribution function. Consequently

$$\frac{\partial}{\partial L}\mathbb{P}(\tau_{MC} < t_1 \mid \mathcal{F}_{t_0}) > 0, \quad \mathbb{P}(\tau_{MC} < t_1 \mid \mathcal{F}_{t_0}) = 1 - \sqrt{\frac{2}{\pi}} \cdot \frac{1 - \alpha}{L\sigma\sqrt{h}} + O(L^{-3}), \quad (18)$$

so that $\mathbb{P}(\tau_{MC} < t_1 \mid \mathcal{F}_{t_0}) \rightarrow 1$ as $L \rightarrow \infty$.

Proof. The result follows from the reflection principle applied to Brownian motion with drift, using the standard first-passage-time density (Shreve, 2004, §3.7.1). Under geometric Brownian motion the log-price barrier corresponding to (16) is

$$b := \ln\left(1 - \frac{1 - \alpha}{L}\right) = -\frac{1 - \alpha}{L} + O(L^{-2}),$$

and the first-passage probability for the downward barrier is

$$\mathbb{P}(\tau_{MC} < h) = \Phi\left(\frac{b - \mu h}{\sigma\sqrt{h}}\right) + e^{2\mu b/\sigma^2} \Phi\left(\frac{b + \mu h}{\sigma\sqrt{h}}\right).$$

Here μ denotes the drift of the log-price process $X_t := \log(S_t/S_{t_0})$ (i.e. $\nu = \mu_S - \frac{1}{2}\sigma^2$ in the notation of Appendix A.2, where μ_S denotes the price drift). Setting $\mu = 0$, the leading-order approximation of a negligible

log-price drift over the short horizons h considered here, and substituting the leading term of b yields (17). Writing $x := (1 - \alpha)/(L\sigma\sqrt{h})$, the scaling in (18) follows from the Taylor expansion $\Phi(-x) = \frac{1}{2} - x/\sqrt{2\pi} + O(x^3)$ valid as $x \rightarrow 0$; here $L \rightarrow \infty$ drives $x \rightarrow 0$ so this is the appropriate expansion (the more familiar Mills-ratio asymptotic $\Phi(-x) \sim \phi(x)/x$ applies as $x \rightarrow \infty$ and would be wrong in this regime). The limit $\mathbb{P} \rightarrow 1$ is intuitive: at very high leverage the maintenance-margin barrier is arbitrarily close to S_{t_0} , so any downward Brownian excursion within horizon h triggers a margin call. $\square \quad \square$

Remark 3.5 (Effect of jumps on the barrier-crossing probability). The diffusion-only expression (17) is not, in general, a strict lower bound on the true Bates margin-call probability. Under the full Bates dynamics (Assumption 3.1) the log-jump-size distribution $\ln J_t \sim \mathcal{N}(\mu_J, \sigma_J^2)$ admits both $J_t > 1$ (upward jumps, which move the price away from the downward barrier) and $J_t < 1$ (downward jumps, which move towards or through it). The net effect of jump inclusion on first-passage probability depends on the calibrated $(\lambda, \mu_J, \sigma_J)$ and the compensator adjustment $-\lambda\kappa$; it is a genuinely two-sided perturbation and must be quantified numerically. The empirical simulation in Section 6 finds that under the illustrative parameterisations adopted, the jump contribution is material at holding horizons above approximately 24 hours. The constant-intensity Poisson process used here does not model event-time clustering (macroeconomic release windows, central-bank announcements); the empirical literature documents elevated jump activity around such events, and modelling event-time clustering via a state-dependent intensity $\lambda(t)$ is left to future work.

3.5. Broker revenue decomposition and B-book internalisation

We decompose the broker's revenue from a single CFD position into three components, following the market microstructure framework of O'Hara (1995), Harris (2003), and Menkveld (2016). This decomposition is essential for the *maysir* analysis of Section 4, which turns on the algebraic relationship between broker P&L and client P&L.

3.5.1. Spread revenue

For a completed round-trip trade of notional N , the broker earns

$$R^{\text{spread}} = N \cdot \frac{s_{t_0} + s_{t_1}}{2}, \quad (19)$$

capturing half the bid–ask spread on entry and half on exit. This component is deterministic conditional on execution and is independent of the price path $\{S_t\}_{t \in [t_0, t_1]}$.

3.5.2. Financing revenue

Aggregate financing revenue over the holding period equals the client’s paid financing:

$$R^{\text{swap}} = \int_{t_0}^{t_1} F_s \, ds, \quad (20)$$

with F_s given by Definition 3.3. This component grows linearly in holding time under stationary interbank rates and constitutes a large fraction of broker revenue on positions held across multiple settlement cutoffs.

3.5.3. Internalisation P&L and the A-book / B-book distinction

Retail brokers operate on a spectrum of internalisation strategies. Let $\beta \in [0, 1]$ denote the fraction of client notional that the broker holds on its own book (“B-book” execution) rather than externalising to an exchange or third-party liquidity provider (“A-book” execution).

- At $\beta = 0$ the broker externalises every client order and earns only spread and financing revenue; its P&L is independent of the client’s directional P&L.
- At $\beta = 1$ the broker takes the opposite side of every client order and holds the resulting exposure. Its P&L on that exposure is the negative of the client’s gross P&L.
- Intermediate β corresponds to selective internalisation, in which the broker retains client flow assessed as unlikely to be informed on its own book and externalises flow assessed as informed; this is the standard practitioner description of retail broker order-management practice, and the associated adverse-selection framework in microstructure is surveyed in the market-maker inventory literature (O’Hara, 1995; Harris, 2003).

Formally, the internalisation P&L component is

$$R^{\text{internal}} = -\beta \cdot \Pi_{\text{gross}}(t_0, t_1), \quad (21)$$

so that the broker's total revenue from a single position is

$$R^{\text{total}} = R^{\text{spread}} + R^{\text{swap}} + R^{\text{internal}} = \underbrace{N \cdot \frac{s_{t_0} + s_{t_1}}{2}}_{\text{spread}} + \underbrace{\int_{t_0}^{t_1} F_s \, ds}_{\text{financing}} - \underbrace{\beta \delta N (S_{t_1} - S_{t_0})}_{\text{internalisation}}. \quad (22)$$

Proposition 3.3 (Zero-sum structure under full B-book). *Suppose the broker operates fully B-book, so that $\beta = 1$, and consider the internalisation $P\mathcal{E}L$ in isolation from spread and financing components. Then the client's gross payoff and the broker's internalisation $P\mathcal{E}L$ sum to zero pathwise:*

$$\Pi_{\text{gross}}(t_0, t_1) + R^{\text{internal}} = 0 \quad a.s., \quad (23)$$

and in particular $\mathbb{E}[\Pi_{\text{gross}}(t_0, t_1)] = -\mathbb{E}[R^{\text{internal}}]$.

Proof. Substitute $\beta = 1$ into (21) and add (3): the sum vanishes identically, pathwise, on Ω . Expectation preservation is then immediate. $\square$ $\square$

Remark 3.6 (Empirical relevance of β). Regulatory disclosure regimes in the European Union (European Securities and Markets Authority, 2018), the United Kingdom (Financial Conduct Authority, 2019), and Australia (Australian Securities and Investments Commission, 2020) require retail CFD brokers to disclose the fraction of client accounts that lose money over the reporting period. The reported loss ranges (74–89%, cited in Section 1.1) are broadly consistent with substantial B-book internalisation but do not identify β : they measure client account outcomes, not the fraction of client notional the broker warehouses on its own book, and external counterparties may also profit systematically from adversely selected retail flow. The paper does not estimate β and does not claim any industry-wide value. Proposition 4.3 below is therefore a conditional structural result: wherever $\beta > 0$, the internalised fraction has the pathwise direct-counterparty zero-sum property. The strength of the *maysir* concern in a specific commercial implementation depends on the value of β that actually obtains there, which is a question of broker order management and outside the scope of this paper.

The zero-sum structure of Proposition 3.3 is the mathematical content of the *maysir* violation formalised in Section 4 as Proposition 4.3. The subject of Section 5 is precisely to construct market-access architectures under which the analogous revenue decomposition satisfies the Shariah constraint set, in

particular, structures under which R^{swap} is not a function of interbank rates and under which R^{internal} is not the pathwise negation of client P&L.

3.6. Primary-source institutional context

The formal characterisation established in Sections 3.1–3.5 draws on public regulatory documentation, the academic literature surveyed in Section 2, and standard market-microstructure references. It is corroborated by primary-source observations gathered during the author’s four-week residency as a Capital Markets and Trading Analyst at Bacera Co Pty Ltd (Sydney, Australia), a global CFD and FX brokerage regulated by the Australian Securities and Investments Commission under Australian Financial Services Licence 328794. The observations recorded here are institutional context and are not relied upon for any specific quantitative claim in the paper; all such claims are grounded in the public sources cited above.

Product mix.. The observed product suite comprised leveraged CFDs on major, minor, and exotic FX pairs; spot metals (XAU/USD, XAG/USD); energy (WTI, Brent); major equity indices (SPX, NDX, DAX, ASX 200, N225); and a subset of individual equities across the US, UK, and Australian markets. Client-facing leverage was capped in accordance with the ASIC 2020 product intervention order for accounts domiciled in Australia: up to 30:1 on major FX pairs, 20:1 on minor FX and gold, 10:1 on non-major indices and commodities, 2:1 on cryptocurrencies, and 5:1 on individual equities.

Financing mechanics.. Tom-Next swaps were applied daily at 5:00 PM New York time (which corresponds to 7:00 AM the following calendar day in Sydney AEDT during Australian summer). A triple-swap charge was applied on Wednesdays, following the T+2 spot settlement convention that shifts the effective rollover date across the weekend on that day of the week. Published swap rates were consistent in sign and order of magnitude with the interbank differential plus markup structure derived formally in equation (8) and were observed to adjust in response to policy actions by the Federal Reserve, the European Central Bank, and the Reserve Bank of Australia during the residency period.

Client outcomes.. Consistent with the published ASIC disclosures cited in Section 1.1, the majority of retail accounts observed during the residency period were in aggregate loss over the reporting horizons for which internal

management dashboards were made available. No individual account-level data are reproduced in this paper.

Islamic account variant. The broker offers a swap-free account variant marketed toward observant-Muslim clients. The variant removes the Tom-Next swap charge for holding periods up to a specified duration (typically between three and fourteen calendar days depending on instrument), after which a fixed administration fee is applied in lieu. Whether this variant resolves the five structural concerns identified in Section 4 is analysed formally in Section 4.7; the answer, foreshadowed, is that it addresses only the first (and only partially), leaving the remaining three structurally intact.

4. Shariah Analysis: Structural Concerns in the Standard CFD Under the Adopted Framework

This section applies the adopted AAOIFI/OIC constraint framework to the formal characterisation of Section 3 and documents five structural incompatibilities or concerns between the standard retail CFD and the operational constraint set adopted in Section 4.1. Section 4.1 formalises the constraint set; Sections 4.2–4.6 state and argue each concern as a formal proposition, invoking the mathematical objects introduced in Section 3; Section 4.7 shows that the “swap-free Islamic account” variant offered by conventional brokers resolves at most one of these concerns, and only partially; and Section 4.8 consolidates the results in a summary table. Throughout the section, the term *violation* is used only where a proposition establishes an unconditional failure under an explicit assumption; where a conclusion is conditional on a scholarly reading (as, e.g., for *maysir* conditional on $\beta > 0$, or *bay’ al-kāli’ bi’l-kāli’* conditional on the classification of margin and broker services), the paper uses *concern*, *structural feature*, or *contested classification* instead.

4.1. The AAOIFI/OIC operational constraint set adopted in this paper

We operationalise the requirements for a bilateral commercial contract to be reviewable as Shariah-consistent by extracting a working constraint set from the primary Accounting and Auditing Organization for Islamic Financial Institutions (2015c) Shariah Standards and the resolutions of the International Islamic Fiqh Academy of the OIC (International Islamic Fiqh Academy, 1992, 2009). The constraint set below is the paper’s operationalisation; it is *not* claimed as a universal necessary-and-sufficient mathematical definition of

Shariah validity in general, which belongs to qualified Shariah supervisory boards and involves additional considerations (contract combination, prohibited underlying activities, procedural conditions, and jurisdictional practice) beyond what a mathematical framework can capture.

Definition 4.1 (Operational Shariah constraint set adopted in this paper). For the purposes of this paper, let $\mathcal{C}_{\text{Shariah}}$ denote the following operational constraint set used to evaluate a bilateral commercial contract $\mathcal{K}$, characterised by the tuple (parties, subject-matter, consideration, terms of exchange), against the selected AAOIFI/OIC framework. A contract is said to be *consistent with $\mathcal{C}_{\text{Shariah}}$ under the operationalisation adopted in this paper* whenever it satisfies each of:

- (C1) *Ridā*, mutual consent: both parties enter the contract freely and with full knowledge of the terms.
- (C2) *Ma'lūmiyya*, the subject matter, the consideration, and the terms of exchange are known and unambiguous at the time of contracting.
- (C3) Absence of *ribā*: no stipulated increment is charged on a loan or extension of credit (*qard* plus *ziyādah*), and no increment on a credit-sale or deferred exchange is priced off a conventional interest-rate benchmark (Usmani, 2002; Kamali, 2000; AAOIFI Shariah Standards No. 1 on Trading in Currencies and No. 8 on *Murābaḥa*, Accounting and Auditing Organization for Islamic Financial Institutions, 2015c).
- (C4) Absence of *maysir*: the contract does not constitute a zero-sum wager whose outcome is determined by chance and to which neither counterparty contributes productive economic activity (OIC Fiqh Academy Resolution No. 63).
- (C5) Absence of excessive *gharar* (*gharar fāḥish*): the uncertainty embedded in the subject matter or the execution is not so great as to render the contract speculative in essence (AAOIFI Standard No. 31; Kamali, 2000).
- (C6) *Qabḍ*: where the contract involves the sale of an asset, the seller has constructive possession of the asset prior to sale (AAOIFI Standard No. 18).
- (C7) Absence of *bay' al-kāli' bi'l-kāli'*: the contract does not constitute the sale of a deferred obligation for a deferred obligation (AAOIFI

Standard No. 8 §2/2; Prophetic prohibition reported in the ḥadīth of Ibn Umar).

Constraints (C1) and (C2) are procedural and, at the level of contract formation, are satisfied by the mutual and voluntary nature of the retail CFD relationship: the client consents to the broker's terms and the terms themselves are documented in the account agreement and product-disclosure statement. The substantive constraints (C3)–(C7) are the object of Propositions 4.1–4.5 below. Denote the standard CFD contract of Definition 3.1 by $\mathcal{K}_{\text{CFD}}$. Under the operationalisation adopted in this paper, this section documents

$$\mathcal{K}_{\text{CFD}} \notin \mathcal{C}_{\text{Shariah}}, \quad (24)$$

via failure of each of (C3), (C4), (C5), (C6), and (C7) in turn. Whether these failures amount to definitive Shariah-non-compliance under a particular scholar's chain of *ijtihād* is a distinct question that this paper does not resolve; the paper's contribution is the formal mathematical mapping from contract mechanics to the operational constraints.

4.2. Ribā al-nasī'ah: the interest-rate financing violation

Proposition 4.1 (*Ribā concern arising from the CFD financing mechanism*). *Let $\mathcal{K}_{\text{CFD}}$ be the standard CFD contract of Definition 3.1 with instantaneous financing rate F_s specified by Definition 3.3. Then $\mathcal{K}_{\text{CFD}}$ fails to satisfy constraint (C3) of Definition 4.1 at the mechanical level: the financing consideration is priced off interbank benchmark rates. Whether this failure amounts to a definitive determination of ribā al-nasī'ah is a matter of legal characterisation for a qualified Shariah Supervisory Board.*

Proof. By Proposition 3.1, the instantaneous financing rate F_s is measurable with respect to the σ -algebra generated by the interbank overnight rates r_s^{CCY1} and r_s^{CCY2} , with unit sensitivity to the differential $r_s^{\text{CCY2}} - r_s^{\text{CCY1}}$ (equation (9)).

The classical characterisation of *ribā al-nasī'ah*, a stipulated fixed increment paid for the deferral of an exchange, over and above the principal (Kamali, 2000, ch. 5; Usmani, 2002, ch. 3), is supported by the CFD architecture through two mechanisms whose combined weight is a matter of scholarly characterisation. *First* (the mechanical concern), the increment F_s is priced off the interbank interest-rate differential (Proposition 3.1); AAOIFI Shariah Standard No. 8 does not permit an increment on a credit or deferred exchange to be linked to a conventional interest-rate benchmark (Accounting

and Auditing Organization for Islamic Financial Institutions, 2015c), a position echoed in the general *ribā* treatment of Usmani (2002) and codified across the subsequent AAOIFI standards. *Second* (the structural characterisation), the leveraged CFD position may be characterised as a credit-like extension of exposure by the broker to the client in nominal amount $(L - 1)/L \cdot N \cdot S_{t_0}$ for the duration $[t_0, t_1]$, on which F_s is a time-proportional periodic charge. Whether that characterisation rises to a *qard* (loan) in the strict fiqh sense, so that the increment becomes definitional *ziyādah*, is a legal question: a CFD broker does not necessarily hand the client cash equal to $(L - 1)/L \cdot N \cdot S_{t_0}$, but grants contractual market exposure against margin. The credit-like-exposure reading is defended in the contemporary Islamic-finance literature (Usmani, 2002; Kamali, 2000) but requires SSB determination in each concrete implementation.

The compliance concern is not resolvable by adjusting the sign of δ , by netting F_s against other cash flows, or by relabelling the fee as a service charge or “financing adjustment”. The mechanical concern (fee priced off interbank rates) follows directly from Proposition 3.1 and requires no additional *ijtihād*. The stronger structural characterisation (leveraged CFD as *qard* plus *ziyādah*) is offered as a defensible fiqh overlay for SSB confirmation, not as a mathematical corollary. The paper therefore documents a mechanical (C3) failure and a structural concern flagged for scholarly assessment. $\square$ $\square$

Remark 4.1 (Consequence for institutional practice). The permissibility of Islamic FX Forwards and Islamic Profit Rate Swaps, offered by Islamic-banking desks at CIMB Islamic, Maybank Islamic, and Standard Chartered Saadiq, is defended in the Bank Negara Malaysia tradition on the grounds that they replace F_s with a *wa’d*-based structure whose consideration is not benchmarked against interbank rates. Under Proposition 4.1, this defence is sound only to the extent that the *wa’d*-based structure’s pricing is genuinely independent of the process r_s^{CCY} . Section 5 constructs such structures explicitly and states the conditions under which the independence is preserved.

Remark 4.2 (Two distinct *ribā* readings). Proposition 4.1 distinguishes a *mechanical* concern (the CFD financing rate F_s is priced off interbank rates, which follows directly from Proposition 3.1) from a *structural* concern (the leveraged CFD may be characterisable as a credit-like exposure whose increment is *ziyādah*). The mechanical concern is a mathematical consequence of the CFD’s financing specification and requires no further *ijtihād*; the structural characterisation is a scholarly overlay defensible in the classical *ribā*

literature (Usmani, 2002; Kamali, 2000) but depends on the legal treatment of the CFD's margin-vs-loan character, which is the province of a qualified SSB. The paper depends only on the mechanical concern for the (C3) failure.

4.3. Gharar fāḥish: the excessive uncertainty violation

Gharar denotes uncertainty inherent to a contract, and *gharar fāḥish* is the threshold at which such uncertainty renders the contract invalid. It is essential to distinguish this concept from ordinary commercial risk: every real commercial transaction, the purchase of a farm, a manufacturing plant, an equity share, involves uncertainty about future value. Such uncertainty is not per se prohibited under Shariah. The threshold for prohibited *gharar fāḥish* is uncertainty that is (a) *material* to the essence of the contract rather than to accidents of its performance; (b) *unbounded or unresolvable* at contract formation, in the sense that essential terms cannot be pinned down; and (c) unaccompanied by a countervailing *commercial function* that justifies bearing the risk. Ordinary business risk fails at least the third leg, the risk-bearing serves the productive purpose of enabling commerce. Under Kamali (2000) this three-part test is operationalised through three factors: (i) uncertainty as to the existence, ownership, or specification of the subject matter; (ii) uncertainty as to the price or consideration; and (iii) uncertainty as to the time or manner of delivery. The CFD is argued below to fail each of these three tests through independent mechanisms; the strength of the resulting classification as *gharar fāḥish* is, ultimately, a fiqh matter that the propositions in this section document rather than resolve.

Definition 4.2 (Execution slippage asymmetry). Let $S_{t_0}^{\text{quote}}$ denote the price displayed on the client's trading terminal at the moment an order is submitted at time t_0 , and $S_{t_0}^{\text{fill}}$ the price at which the order is actually executed by the broker's matching engine. The *execution slippage* is

$$\Xi_{t_0} := \delta \cdot (S_{t_0}^{\text{fill}} - S_{t_0}^{\text{quote}}), \quad (25)$$

adopting the sign convention that $\Xi_{t_0} > 0$ is adverse to the client. Execution slippage is *asymmetric against the client* if $\mathbb{E}[\Xi_{t_0}] > 0$.

Proposition 4.2 (*Gharar fāḥish* concern). *Let $\mathcal{K}_{CFD}$ be the standard CFD contract of Definition 3.1. Then $\mathcal{K}_{CFD}$ exhibits three independent structural features that together constitute a substantial gharar fāḥish concern under the*

framework of Definition 4.1: (a) exposure to a synthetic contingent payoff without ownership, delivery, or commercial anchoring; (b) forced-liquidation risk under leveraged maintenance-margin mechanics; and (c) execution ambiguity through asymmetric slippage between quoted and filled prices.

Proof. Source (a): synthetic contingent payoff without ownership. By Remark 3.3, the client acquires no rights in the underlying and no commercial exposure other than a claim to the future price difference $S_{t_1} - S_{t_0}$. The subject matter of the contract is therefore a synthetic contingent payoff, not an underlying commodity, currency, or equity. Under Kamali (2000), the classical *gharar* threshold turns not on nonzero variance *per se* (future prices of farms, businesses, and any real asset are uncertain) but on the conjunction of (i) contractual uncertainty as to the essence of the subject matter, (ii) absence of underlying ownership or delivery, and (iii) absence of a commercial function that anchors the risk-bearing to genuine economic activity. The CFD structure satisfies (i)–(iii) simultaneously: the subject matter is a scalar $S_{t_1} - S_{t_0}$ rather than a delivered asset; no ownership of the underlying is transferred at any point (Remark 3.3); and the contract does not involve the acquisition, production, or use of the underlying in commerce. Under Assumption 3.1, the payoff $S_{t_1} - S_{t_0}$ has strictly positive conditional variance,

$$\mathbb{V}[S_{t_1} - S_{t_0} \mid \mathcal{F}_{t_0}] > 0, \quad (26)$$

with support on $(-S_{t_0}, +\infty)$; but the paper’s *gharar* argument rests on the conjunction (i)–(iii) rather than on the positive variance alone. This is the configuration that AAOIFI Standard No. 31 and the Kamali three-factor test identify as characteristic of prohibited *gharar* in a bilateral commercial contract.

Source (b): termination-time uncertainty from maintenance-margin close-out. By Proposition 3.2, the client’s realised holding horizon is not entirely within the client’s control: with positive probability $\mathbb{P}(\tau_{\text{MC}} < t_1) > 0$, the position is closed out at τ_{MC} , a stopping time whose realisation is not observable at contract inception t_0 . This creates a termination-time uncertainty. A cautious scholarly reading is that this is not the same as classical uncertainty about the time or manner of *delivery of the subject matter* (a scholar could reasonably object that the contract specifies the maintenance-margin trigger clearly, and the uncertainty concerns only whether that trigger occurs, not what is being delivered). The paper therefore treats source (b) as a

termination-time uncertainty contributing to the broader *gharar* concern rather than as a self-sufficient demonstration of Kamali (2000)'s third factor; the combination with sources (a) and (c) supplies the substantive weight of the (C5) concern.

Source (c): execution ambiguity through asymmetric slippage. Definition 4.2 makes precise the notion of asymmetric slippage $\Xi = \delta(S^{\text{fill}} - S^{\text{quote}})$ that arises in retail CFD execution. The proposition depends on the analytical condition $\mathbb{E}[\Xi] > 0$; that condition is motivated by, but not directly estimated from, the retail execution-quality material regulators have published (e.g. Financial Conduct Authority, 2019, European Securities and Markets Authority, 2018, which document adverse execution outcomes, rejected orders and price-improvement asymmetries without publishing the expectation of the exact Ξ -statistic defined here). The condition $\mathbb{E}[\Xi] > 0$ is therefore introduced as an analytical assumption consistent with the direction of that regulatory evidence, not as an empirically estimated industry parameter. Under the assumption, the client cannot observe Ξ_{t_0} at order submission and bears uncertainty as to the true price of the transaction, the second of Kamali (2000)'s three factors.

Each of sources (a), (b), (c) documents a structural feature that the AAOIFI Standard No. 31 and OIC Fiqh Academy Resolution No. 63 frameworks identify as characteristic of *gharar fāḥish*. Their combination *a fortiori* constitutes an argument that the CFD fails constraint (C5). Whether this argument is dispositive under classical fiqh criteria is a scholarly determination outside the scope of the present paper; the propositions above establish the structural conditions on which such a determination would rest. $\square \quad \square$

4.4. Maysir: the zero-sum wager violation

Proposition 4.3 (Conditional *maysir* argument under B-book execution). *Let $\mathcal{K}_{CFD}$ be the standard CFD contract of Definition 3.1 with internalisation ratio $\beta > 0$. Then, on the internalised fraction $\beta \cdot N$ of client notional, $\mathcal{K}_{CFD}$ satisfies the structural conditions for maysir under International Islamic Fiqh Academy (1992). The overall compliance verdict under constraint (C4) is discussed in Remarks 4.3–4.4 below.*

Proof. By Proposition 3.3, under $\beta = 1$ the client's gross P&L and the broker's internalisation P&L sum to zero pathwise:

$$\Pi_{\text{gross}}(t_0, t_1) + R^{\text{internal}} = 0 \quad \text{a.s.} \quad (27)$$

For intermediate $\beta \in (0, 1)$, the same identity holds on the internalised fraction $\beta \cdot N$ of the notional exposure. On that internalised fraction, the transaction is a direct and pathwise mutually exclusive transfer of wealth between the two identified counterparties, whose outcome is determined by the future realisation of the price process $(S_t)_{t \geq 0}$, over which neither counterparty exercises control.

The paper operationalises *maysir* using the following three-element structural test, informed by IIFA Resolution No. 63 (International Islamic Fiqh Academy, 1992, which discusses financial-market structures and characterises certain cash-settled derivative constructions, including index trading, as impermissible on the ground that they constitute pure gambling on a fictitious subject-matter) and the contemporary treatments of Kamali (2000), Usmani (2002) and El-Gamal (2006): a transaction exhibits the structural features of *maysir* when (i) wealth transfers directly and mutually exclusively between identified counterparties, (ii) the transfer is contingent on a future event whose outcome is uncertain at inception, and (iii) neither counterparty contributes productive economic activity to the outcome. The internalised component of the CFD satisfies each element: (i) the transfer is direct and mutually exclusive between the two identified counterparties (Prop. 3.3); (ii) it depends on $S_{t_1} - S_{t_0}$, unresolved at t_0 ; (iii) neither counterparty contributes to the productive activity underlying S (Remark 3.3). Hence, on the internalised fraction, the structural test above is satisfied; whether this is dispositive of *maysir* under a specific SSB reading is a matter outside the paper's scope. $\square \quad \square$

Remark 4.3 (Beyond B-book: the counterparty-agnostic reciprocity argument). Proposition 4.3 is deliberately conditional on $\beta > 0$: the pathwise zero-sum identity of Proposition 3.3 rests on the broker being the client's direct counterparty for the internalised notional. A separate, counterparty-agnostic, argument may be constructed: the CFD is a cash-settled bilateral contract in which for every dollar the client gains, some counterparty, broker (B-book), external liquidity provider (A-book), or mixture thereof, loses that dollar, because no productive real-world activity generates the P&L. Whether this counterparty-agnostic reciprocity itself constitutes *maysir* under classical criteria is a distinct scholarly question. Some contemporary scholars (Usmani, 2002; El-Gamal, 2006) treat the reciprocity as sufficient regardless of the identity of the counterparty; others restrict the classification to direct-counterparty structures. This paper proves only the direct-counterparty case; the broader argument is noted as a scholarly extension awaiting SSB determination.

Remark 4.4 (Empirical uncertainty about β). Any claim that retail CFD brokers exhibit large β values is *inferential* rather than directly measured: regulatory disclosures report the fraction of client accounts that lose money, not the internalisation ratio. A large disclosed loss rate is consistent with substantial B-book internalisation but does not identify β uniquely, external counterparties may also profit systematically from adversely selected retail flow, a possibility surveyed in the retail-microstructure literature (Barber and Odean, 2000). The empirical claim underlying the (C4) analysis is therefore only that β is materially positive for the retail segment; the proposition above is careful to depend solely on this qualitative claim rather than on any specific point estimate.

Remark 4.5 (Distinction from commercial hedging). A common defence of derivative instruments in Islamic finance is that they may permissibly discharge a hedging function: a farmer using a futures contract to hedge a harvest, an exporter using an FX forward to hedge foreign-currency receivables. This defence is inapplicable to the retail CFD use case, in which the participant has no underlying commercial exposure to hedge; the CFD position *is* the exposure. See Section 5 for treatment of the genuine hedging use case under a *wa'd*-based framework consistent with the Bacha and Mirakhor (2013) tradition.

4.5. The contract-classification problem

An objection to the standard *qabḍ* argument is that AAOIFI Standard No. 18's constructive-possession requirement applies specifically to contracts of sale (*bay'*); a respondent might argue that a cash-settled CFD is not a *bay'* of the underlying at all, and that the *qabḍ* requirement is therefore misapplied. Proposition 4.4 below reframes the analysis around the more fundamental question of *contract classification*: the CFD, on inspection, does not correspond to any classical nominate contract. Non-fit with the nominate contracts is not by itself a determination of impermissibility (classical fiqh admits new contract forms subject to the substantive prohibitions); it means the substantive prohibitions of *ribā*, *gharar*, and *maysir* apply directly to the novel contract without the mediating discipline of a recognised nominate category, which is the deeper scholarly concern.

Proposition 4.4 (Contract-classification: the CFD is not any classical nominate contract). *The standard CFD contract of Definition 3.1 does not correspond to any of the classical nominate contract categories enumerated*

in *Accounting and Auditing Organization for Islamic Financial Institutions* (2015c, Sec. 2.1) and reviewed in Section 2.1: it is not bay' (no qabḍ of the underlying; no transfer of ownership rights characteristic of a sale); not salam (no immediate full payment of price against a deferred fungible commodity); not istiṣnā' (no commissioning of manufacture); not sharīkah (no capital pooling for joint ownership); not ijāra (no lease of usufruct); not wakāla (no defined agency for a service fee); and not wa'd (no unilateral binding promise). It is, structurally, a novel contract of cash-settled contingent obligations. Under the general principles surveyed by Kamali (2000), Usmani (2002), and El-Gamal (2006), contracts falling outside the recognised nominate categories are evaluated against the substantive prohibitions of ribā, gharar, and maysir, the analyses of Sections 4.2–4.4 above.

Proof. Verification of non-fit is by inspection of Definitions 3.1 and 3.3 against the classical categories catalogued in the literature review. The CFD confers no ownership of, or legal claim to, N units of the underlying (Remark 3.3); it requires no upfront payment of the notional (only initial margin M_0); it involves no manufacture, lease, agency, partnership, or unilateral promise. The residual category is that of cash-settled contingent obligation, which is not a classical nominate category. Under the general-principles fallback of Kamali (2000), the substantive prohibitions apply directly, which is the subject of Sections 4.2–4.4. $\square$ $\square$

Remark 4.6 (On the classical *qabḍ* formulation). Proposition 4.4 is a reformulation of the more familiar claim that the CFD violates the *qabḍ* requirement of AAOIFI Standard No. 18. That familiar claim assumes, implicitly, that the CFD is (or purports to be) a *bay'* of the underlying, and then observes that *qabḍ* is absent. The present reformulation avoids the assumption: rather than asking whether the CFD satisfies the *qabḍ* requirement of a *bay'*, it asks whether the CFD is a *bay'* in the first place. The answer is no, and also no to every other nominate category. Constraint (C6) is therefore best understood not as a violation of *qabḍ* in isolation but as a symptom of the deeper non-fit problem, which is itself the foundational scholarly concern.

4.6. Bay' al-kālī' bi'l-kālī': the deferred-for-deferred concern

Proposition 4.5 (Deferred-for-deferred structural feature). *The standard CFD contract of Definition 3.1 with $t_0 < t_1$ exhibits the reciprocal-deferred-contingent structure that the classical prohibition of bay' al-kālī' bi'l-kālī' addresses.*

Proof sketch of the structural feature. The Prophetic prohibition against *bay' al-kāli' bi'l-kāli'*, reported in the ḥadīth of Ibn Umar and codified in AAOIFI Shariah Standard No. 8 §2/2, forbids the exchange of one deferred obligation for another. Under the reading in which the initial margin M_0 is classified as a security deposit rather than as consideration, and the reading in which broker execution, platform access and counterparty performance do not themselves constitute immediate counter-consideration, the following structural feature is present. At contract inception t_0 , the client posts only $M_0 = N \cdot S_{t_0}/L$ (equation (11)), which under the assumed reading is a security deposit held against margin-call events; the broker under the same reading delivers no goods, no cash, and no asset at t_0 . Both parties' substantive obligations, settlement of $\Pi_{\text{net}}(t_0, t_1)$ per equation (4), are deferred to t_1 and are moreover contingent on the terminal state S_{t_1} . Neither party under the assumed reading takes immediate *qabd* (Proposition 4.4) or renders immediate counter-consideration. Under those readings, the reciprocal-deferred-contingent structural feature that the classical prohibition addresses is present. Each reading is plausible and each is contested: a scholar could take the alternative view (that margin is partial consideration; that broker services constitute immediate counter-consideration), in which case the structural feature would not be present in the same form. The paper documents the structural concern under the broad reading and leaves the definitive classification to SSB review. $\square$ $\square$

Remark 4.7 (Scope of the classical prohibition). The classical prohibition of *bay' al-kāli' bi'l-kāli'* addresses, at its narrowest reading, the sale of a debt for a debt. Its extension to non-sale contracts exhibiting the same reciprocal-deferred structure is a matter of scholarly extension: Usmani (2002) and the majority of GCC scholars take the broad view, extending the prohibition wherever both counter-values are deferred; a subset of Malaysian scholars restrict the classical prohibition to formal sales. Combined with the non-fit result of Proposition 4.4, however, the CFD is on any reading outside the class of contracts recognised as permissible under classical fiqh: it is neither a permitted deferred sale (e.g. *salam*, in which the price is paid in full at inception) nor a permitted spot sale (in which both counter-values are exchanged immediately). Whether its exclusion is best framed as violation of (C7) narrowly or as violation of the general non-fit principle of Proposition 4.4 is a scholarly question; the paper records the structural feature and both framings.

Remark 4.8 (The margin-as-consideration defence and the broker-service defence). Two defences may be raised against Proposition 4.5. The *margin-as-consideration defence* holds that M_0 constitutes immediate consideration by the client; against this, M_0 is fully returnable to the client at t_1 absent adverse price movement of magnitude exceeding $(1 - \alpha)M_0$, which is more characteristic of a security deposit than of a payment for goods or services. The *broker-service defence* holds that the broker’s execution, platform access, counterparty performance, and liquidity provision are themselves immediate counter-consideration; against this, these services are not the primary subject-matter of the CFD (which is the price-differential settlement at t_1) and are in any event provided against separate transaction fees where they are charged. Neither defence is definitively refuted; both are plausible scholarly positions the paper acknowledges. The paper documents the structural concern that survives *if* one takes the broad reading of the classical prohibition together with the assumed reading of margin and broker services, and leaves the definitive classification to SSB review.

4.7. The “Islamic account” variant: a formal analysis

We now analyse whether the swap-free “Islamic account” variant offered by conventional brokers (Section 3.6) resolves the five violations proved in Sections 4.2–4.6.

Definition 4.3 (Swap-free variant). The swap-free Islamic-account variant $\mathcal{K}_{\text{CFD-SF}}$ modifies the standard CFD contract $\mathcal{K}_{\text{CFD}}$ by replacing the Tom-Next financing rate of Definition 3.3 with

$$\tilde{F}_s = \begin{cases} 0 & \text{if } s - t_0 \leq \Delta_{\text{grace}}, \\ \phi_{\text{admin}} & \text{if } s - t_0 > \Delta_{\text{grace}}, \end{cases} \quad (28)$$

where Δ_{grace} is a broker-specified grace period (typically in the range 3–14 calendar days) and $\phi_{\text{admin}} > 0$ is a fixed per-diem administrative fee accruing beyond the grace window. All other elements of the contract tuple $(t_0, t_1, \delta, N, L, s, m)$ are unchanged.

Proposition 4.6 (Structural insufficiency of the swap-free variant). *The swap-free variant $\mathcal{K}_{\text{CFD-SF}}$ resolves constraint (C3) partially, only for holding periods $s - t_0 \leq \Delta_{\text{grace}}$, and leaves constraints (C4), (C5), (C6), and (C7) structurally unresolved.*

Proof. (i) Constraint (C3). For $s - t_0 \leq \Delta_{\text{grace}}$, the modified financing rate $\tilde{F}_s = 0$ carries no interest-rate dependence and Proposition 3.1 does not apply. For $s - t_0 > \Delta_{\text{grace}}$, the flat fee ϕ_{admin} may or may not violate (C3) depending on whether it is calibrated against prevailing interbank rates in the broker's pricing model. Brokers do not typically disclose this calibration, and the scholarly consensus is that flat administrative fees are permissible only if their magnitude is not indexed to interest-rate benchmarks either directly or by regular recalibration (Al-Suwailem, 2006; Uberoi and Evans, 2008). The (C3) resolution is therefore partial and conditional.

(ii) Constraint (C4). The internalisation ratio β and the pathwise zero-sum relation of Proposition 3.3 are independent of the specification of the financing rate. Proposition 4.3 applies verbatim to $\mathcal{K}_{\text{CFD-SF}}$. The *maysir* violation is unresolved.

(iii) Constraint (C5). The three sources of *gharar* identified in Proposition 4.2, unbounded terminal price uncertainty, forced-liquidation risk, and execution ambiguity, are all independent of the financing rate. The *gharar fāḥish* violation is unresolved.

(iv) Constraint (C6). The absence of *qabḍ* (Proposition 4.4) is a property of the CFD contract structure itself and is independent of the specification of $\tilde{F}_s$. Unresolved.

(v) Constraint (C7). The *bay' al-kāli' bi'l-kāli'* structure (Proposition 4.5) is likewise independent of the financing rate. Unresolved. $\square$ $\square$

Remark 4.9 (Marketing versus structural compliance). The swap-free variant is therefore best understood as a marketing product that addresses the most conspicuous fiqh objection, interest-linked financing, while leaving the deeper structural objections intact. Its usage may be defended only under a chain of *ijtihād* that treats the *qabḍ*, *maysir*, *gharar*, and *kāli'-bi'l-kāli'* requirements as non-operative for cash-settled derivative contracts, a position that is not the one adopted in the AAOIFI/OIC framework used in this paper, nor in the classical fiqh treatments of Kamali (2000), El-Gamal (2006), and Usmani (2002). Alternative permissive positions on cash-settled derivative structures exist in the Bank Negara Malaysia and ISRA scholarly tradition (see Section 5.2 and Remark 5.1) and remain the subject of active scholarly disagreement.

4.8. Summary of structural violations

The results of Sections 4.2–4.7 are consolidated in Table 1. The table records, for each of the five substantive Shariah constraints, the proposition

establishing the violation, the primary AAOIFI Standard or OIC resolution invoked, and the compliance status under the standard retail CFD (Definition 3.1), the swap-free variant (Definition 4.3), and each of the three alternative architectures constructed in Section 5.

Table 1: Documented Shariah concerns for retail CFD products and their treatment across contract variants. Entries describe the paper’s characterisation of each concern under the standard CFD, the “swap-free Islamic account” variant (SF variant), and the *sharikat al-’inān* alternative architecture of Section 5. All classifications are provisional and subject to review by qualified Shariah supervisory boards.

Concern	Primary source	Argued in	CFD	SF variant	Alt. arch.
(C3) <i>ribā</i>	AAOIFI SS 1, 8; Usmani, Kamali	Prop. 4.1	structural concern	partially addressed	satisfied by design
(C4) <i>maysir</i>	OIC Res. 63 (1/7)	Prop. 4.3	conditional on $\beta > 0$	unchanged	satisfied by design
(C5) <i>gharar</i>	AAOIFI SS 31; Kamali	Prop. 4.2	documented concern	unchanged	largely mitigated
(C6) <i>qabḍ</i> / non-fit	AAOIFI SS 18	Prop. 4.4	non-fit with nominate contracts	unchanged	joint ownership from t_0
(C7) <i>kālī</i>	AAOIFI SS 8	Prop. 4.5	structural concern	unchanged	simultaneous consideration

Table 1 makes the paper’s central negative finding precise: the standard retail CFD violates five foundational Shariah constraints, and the “swap-free Islamic account” variant offered by conventional brokers resolves at most one of these, and only partially. The task of Section 5 is to construct market-access architectures that resolve all five simultaneously, and Section 6 quantifies the economic cost of doing so.

5. First-Principles Alternative Architectures

Sections 3–4 established that the standard retail CFD violates five foundational constraints of Islamic commercial law and that the swap-free variant marketed by conventional brokers resolves at most one of these, and only partially. This section undertakes the constructive task of the paper: to design

market-access architectures that satisfy all five constraints simultaneously and to derive their payoff dynamics under the same jump-diffusion process (Assumption 3.1) used for the standard CFD.

Section 5.1 restates the constraint set of Section 4 as design requirements on any candidate architecture. Sections 5.2–5.4 introduce and analyse three architectures, *wa'd*-based FX exposure, organised *tawarruq* for financing, and *sharīkat al-'inān* pooled-equity market access, with the third being the paper's principal constructive contribution. Section 5.5 consolidates the architectures in a comparative table. Section 5.6 acknowledges the retail CFD use cases for which no permissible equivalent exists.

5.1. Design constraints on a compliant architecture

The Shariah constraint set $\mathcal{C}_{\text{Shariah}}$ of Definition 4.1 translates, at the level of contract engineering, into five design conditions on any candidate market-access architecture.

Definition 5.1 (Operational design constraints adopted in this paper). The design conditions (D1)–(D5) below are the operational constraints adopted in this paper for evaluating candidate market-access architectures against the framework of Definition 4.1. They are not claimed as necessary and sufficient conditions for Shariah compliance in general (a claim which properly belongs to qualified Shariah supervisory boards and involves additional considerations including agency, custody, contract combination, prohibited underlying activities, contractual conditions, and settlement rights). They are a researcher's working framework that operationalises the five concerns of Definition 4.1 in a form amenable to mathematical characterisation.

- (D1) *No interest-linked consideration.* The periodic financing rate $F_s^\mathcal{K}$, if any, has no functional dependence on any interbank rate process. Let $\mathcal{G}_s^\mathcal{K}$ denote the σ -algebra generated by the underlying price and any other non-rate variables that may enter $F_s^\mathcal{K}$; the formal requirement is that $F_s^\mathcal{K}$ is $\mathcal{G}_s^\mathcal{K}$ -measurable, or equivalently $\partial F_s^\mathcal{K} / \partial r_s^{\text{CCY1}} = \partial F_s^\mathcal{K} / \partial r_s^{\text{CCY2}} = \partial F_s^\mathcal{K} / \partial r_s^{\text{IBOR}} = 0$ holds jointly. Addresses (C3).
- (D2) *Constructive possession (qabḍ) where a sale is present.* For any contract element construed as a sale of an underlying asset, at least one counterparty holds legal or constructive ownership of the traded quantity prior to sale. Addresses (C6).

- (D3) *Real-economic-activity linkage*. The payoff derives from the acquisition, production, transformation, or delivery of an identifiable real asset, not from cash-settled price differences alone. Addresses (C4) and (C7).
- (D4) *Proportional risk sharing*. Where the architecture is constructed as a partnership, *losses* must be allocated strictly in proportion to capital contribution (AAOIFI Standard No. 12 §3/1/5/2); *profits* may be distributed under the pre-agreed permissible profit-sharing ratio, which need not equal the capital ratio. Addresses (C4).
- (D5) *Immediate substantive consideration where required*. At contract inception, at least one counterparty renders immediate non-contingent consideration in circumstances where the applicable contract category so requires. Addresses (C7).

The remaining substantive concern, excessive *gharar* (C5), is treated as a parameter constraint rather than a binary design condition: candidate parameterisations that introduce forced-liquidation risk on the client, unbounded terminal exposure without commercial backing, or execution ambiguity in the sense of Definition 4.2 are flagged for scrutiny rather than automatically excluded.

Sections 5.2–5.4 evaluate three candidate architectures against Definition 5.1, recording explicitly where each falls short of one or more of (D1)–(D5) under alternative scholarly readings.

5.2. Architecture A: Wa'd-based FX exposure

The *wa'd*-based FX exposure architecture is a well-documented Shariah-compliant substitute for the conventional FX forward and FX swap in institutional Islamic banking. It has been permitted by the Shariah Advisory Council of Bank Negara Malaysia since 2005 (Bacha and Mirakhor, 2013) and forms the basis of the Islamic FX Forward and Islamic Profit Rate Swap products offered by a number of Malaysian, Middle Eastern, and international Islamic banking desks; comprehensive market-share statistics for the Islamic derivatives universe are not publicly available. It is contested by a substantial portion of Gulf Cooperation Council scholars, whose objections we record honestly.

Definition 5.2 (*Wa'd-based FX exposure*). The *wa'd*-based FX exposure architecture $\mathcal{K}_{wa'd}$ on the FX pair CCY1/CCY2 comprises the following.

At contract inception t_0 :

1. Client A issues a unilateral binding promise (*wa'd mulzim*) to purchase N units of CCY1 from Broker B at the pre-agreed rate K at delivery time T .
2. Broker B issues a corresponding unilateral binding promise to sell N units of CCY1 to Client A at rate K at delivery time T .

At delivery time T :

3. Both parties execute a spot FX transaction: A pays $N \cdot K$ units of CCY2 to B , and B delivers N units of CCY1 to A . Both parties take immediate *qabḍ* of the exchanged currencies.

The client may thereafter sell the received N units of CCY1 at the prevailing spot rate S_T , for a net cash outcome in CCY2 as in Proposition 5.1 below.

Proposition 5.1 (*Wa'd payoff dynamics under Assumption 3.1*). *Under Definition 5.2 and Assumption 3.1 on the underlying spot process (S_t) , the client's terminal cash outcome in units of CCY2, net of the broker's structuring fee $u^{wa'd} \geq 0$, is*

$$\Pi_c^{wa'd}(t_0, T) = N \cdot (S_T - K) - u^{wa'd}. \quad (29)$$

The value at $t \in [t_0, T]$ is $\Pi_c^{wa'd}(t) = N \cdot (S_t - K) e^{-r(T-t)} - u^{wa'd} e^{-r(T-t)}$ under the risk-neutral measure.

Proof. By construction of Definition 5.2, both promises are executed at T at rate K : the client receives N units of CCY1 in exchange for $N \cdot K$ units of CCY2. Immediate sale of the N units of CCY1 at the prevailing spot S_T yields $N \cdot S_T$ units of CCY2. Netting the two cash flows and the *ujrah* gives (29). Measurability of $\Pi_c^{wa'd}$ with respect to (S_t) is immediate; no dependence on interbank rates enters through the transaction structure. The time- t value is a standard risk-neutral valuation (Björk, 2009, ch. 7). $\square$ $\square$

Cost of carry: wa'd versus conventional NDF. Under the risk-neutral measure, the expected terminal spot equals the forward implied by covered interest rate parity, $\mathbb{E}^*[S_T] = F_0^{\text{CIP}} = S_{t_0} e^{(r^{\text{CCY2}} - r^{\text{CCY1}})(T-t_0)}$ (Björk, 2009, ch. 24). If the *wa'd* strike is set to $K = F_0^{\text{CIP}}$, the expected payoff net of *ujrah* equals zero,

economically equivalent to a conventional non-deliverable forward (NDF). The scholarly debate turns on precisely this point: if K is calibrated to the CIP-implied forward rate, the *wa'd* structure inherits an economic dependence on interbank rates even though its contractual mechanics do not reference them. Remark 4.1 foreshadowed this concern.

Remark 5.1 (The scholarly debate on the double-*wa'd* structure). The most serious concern with Definition 5.2 is not (C3) but the more fundamental question of whether a *bilateral* binding *wa'd* structure, both parties issuing binding promises against each other, is permissible at all. AAOIFI Shariah Standard No. 1 draws an express distinction between a single binding unilateral promise (which may be permissible under specified conditions) and a binding bilateral promise (*muwā'adah*) to purchase and sell currencies against each other (Accounting and Auditing Organization for Islamic Financial Institutions, 2015c). AAOIFI Shariah Standard No. 49 (Accounting and Auditing Organization for Islamic Financial Institutions, 2015b) provides the standards-body's dedicated treatment of the distinction and of the conditions under which each may operate. The International Islamic Fiqh Academy Resolution No. 238 (International Islamic Fiqh Academy, 2019) discusses mutual binding commitments in future currency exchange along similar lines. Under the restrictive reading of these authorities, Definition 5.2, which requires both counterparties to issue binding promises, would not be permissible even before the (C3) pricing question arises.

The permissive tradition associated with the Bank Negara Malaysia Shariah Advisory Council and ISRA scholarship is more nuanced than a blanket endorsement of double-*wa'd* structures. BNM's earlier rulings on forward FX permitted the use of a *unilateral* binding *wa'd* by one counterparty and, notably, took the position that no fee could be charged on the promisee for the promise itself (a fee on the promise being one of the features that converts a unilateral into a de facto bilateral commitment). Later BNM rulings addressed *muwā'adah* more directly and admitted certain bilateral structures subject to specified governance conditions. The exact architecture of Definition 5.2, a double binding *wa'd* plus a structuring fee, should not therefore be assumed to inherit any of these BNM rulings automatically; each specific implementation would need SSB assessment under whichever authority governs it. The *restrictive* position associated with Usmani (2002), El-Gamal (2006), and the majority of GCC scholars treats such bilateral binding structures as a form of *hiyal* that reproduces the economic substance

of a forward contract. This paper adopts the restrictive AAOIFI reading; the divergence is recorded, and consequently Definition 5.2 is *not* presented as an unconditionally compliant architecture.

A separate concern arises with the structuring fee $u^{\text{wa'd}} \geq 0$. Both AAOIFI SS No. 49 (Accounting and Auditing Organization for Islamic Financial Institutions, 2015b, in its general treatment of promise-based structures) and IIFA Resolution No. 238 (International Islamic Fiqh Academy, 2019) disfavour consideration paid for the binding commitment itself; a fee-on-promise is one of the features that transforms a unilateral promise into an effectively bilateral one. The paper's treatment of $u^{\text{wa'd}}$ therefore requires that the fee be clearly identified, in a separately executed management-services agreement, as consideration for a distinguishable service (execution, documentation, custody), never as payment for the promise itself; whether that separation is respected in practice is a matter for SSB review of any specific implementation.

Proposition 5.2 (Contested compliance of $\mathcal{K}_{\text{wa'd}}$). *Under Definition 5.2, the compliance status of $\mathcal{K}_{\text{wa'd}}$ with the design constraints (D1)–(D5) of Definition 5.1 is contested:*

- (a) *The bilateral binding-promise structure of clauses 1–2 of Definition 5.2 sits in tension with the AAOIFI restrictive reading of Standard No. 1 on muwā'adah in currencies, and with the IIFA Resolution No. 238 position; under those authorities the architecture is not permissible in its stated form.*
- (b) *Under the permissive reading (Bank Negara Malaysia Shariah Advisory Council), the architecture may be permissible subject to governance conditions on the unilateral character of each promise, actual delivery at T , and separately identifiable service consideration for $u^{\text{wa'd}}$ per Remark 5.1.*
- (c) *Even on the permissive reading, (D1) is respected only if the strike K and the fee $u^{\text{wa'd}}$ are set independently of the interbank-rate processes; calibration to the CIP-implied forward F_0^{CIP} reintroduces the concern of Proposition 3.1.*

Proof sketch. Clauses (a)–(c) follow by inspection of Definition 5.2 against, respectively, the AAOIFI SS No. 1 restriction on binding bilateral currency promises, the Bank Negara Malaysia permissive framework, and Proposition 3.1. The paper does not adjudicate between the permissive and restrictive

readings; both are recorded, and consequently the architecture is not carried forward in this paper as a strictly recommended option. $\square$ $\square$

Remark 5.2 (Consequence for the paper). Because Architecture A is contested at the fundamental level (not only in its calibration under (D1)), this paper’s constructive contribution is centred on Architecture C (*sharīkat al-’inān*), Section 5.4. Architecture A is retained as an intermediate reference against which the constructive contribution is contrasted, not as a standalone recommendation.

Proof sketch. (C4): the transaction is not zero-sum, both parties genuinely exchange currencies to their commercial benefit; there is no mutually exclusive transfer conditional on price. (C5): the contract terms and delivery are fixed at t_0 ; there is no forced-liquidation risk in the sense of Proposition 3.2. (C6): both parties take immediate *qabḍ* at T . (C7): the client’s payment and the broker’s delivery at T are simultaneous, not both deferred. (C3): the contract mechanics of Definition 5.2 do not reference interbank rates, so (C3) is satisfied at the mechanical level; but if K or $u^{\text{wa'd}}$ is calibrated to F_0^{CIP} , the economic exposure inherits interbank dependence and (C3) is violated at the calibration level. $\square$ $\square$

5.3. Architecture B: Organised *tawarruq* financing overlay

The *tawarruq* structure synthesises cash financing through a coordinated sequence of commodity transactions. It is included for completeness, organised *tawarruq* is widely deployed across Middle Eastern and Malaysian Islamic financing desks, though without publicly available market-share statistics, but is *not* recommended by this paper, on the grounds of the scholarly consensus recorded in Remark 5.3.

Definition 5.3 (Organised *tawarruq* financing overlay). The organised *tawarruq* financing overlay $\mathcal{K}_{\text{taw}}$ comprises the following sequence.

At inception t_0 :

1. Bank purchases a quantity Q of a fungible commodity, typically London Metal Exchange copper or aluminium warrants, at spot price P_c per unit.
2. Bank sells the Q units to the client at the deferred price $P_d = P_c \cdot (1 + \mu)$ per unit, payable at maturity t_1 , where $\mu \geq 0$ is the *murābaḥa* markup.
3. Client sells the Q units immediately at spot to a third-party counterparty at price $P_r \leq P_c$ per unit, receiving cash $Q \cdot P_r$.

At maturity t_1 :

4. Client pays $Q \cdot P_d$ to the bank in cash.

Proposition 5.3 (*Tawarruq financing cost*). *The client's net financing cost over $[t_0, t_1]$ under Definition 5.3 is*

$$C^{taw}(t_0, t_1) = Q \cdot (P_d - P_r) = Q \cdot P_c \cdot (1 + \mu) - Q \cdot P_r. \quad (30)$$

The effective annualised cost of the synthesised cash financing $Q \cdot P_r$ is

$$r^{eff} = \frac{Q \cdot (P_d - P_r)}{Q \cdot P_r \cdot (t_1 - t_0)} = \frac{P_c(1 + \mu) - P_r}{P_r \cdot (t_1 - t_0)}. \quad (31)$$

Proof. Direct from the cash flows of Definition 5.3: the client receives $Q \cdot P_r$ at t_0 and pays $Q \cdot P_d$ at t_1 . The net cost is the difference; the effective rate follows by dividing the cost by the principal and the tenor. $\square$ $\square$

Remark 5.3 (Scholarly criticism of organised *tawarruq*). The International Islamic Fiqh Academy of the OIC has explicitly ruled that organised *tawarruq*, in which the bank coordinates all four transactions and the third-party purchaser is pre-arranged, is impermissible (International Islamic Fiqh Academy, 2009). Usmani (2002) and the majority of contemporary GCC scholars have taken the same position, on the ground that the structure is *hiyal* designed to synthesise interest-bearing financing. The practical usage of organised *tawarruq* in Islamic banking is defended by some Malaysian scholars and, on necessity grounds, by Al-Suwailem (2006); but the consensus AAOIFI and OIC position remains restrictive. In institutional practice, the markup μ is almost invariably calibrated against a prevailing interbank benchmark, which additionally invokes the (C3) concerns of Proposition 4.1.

Proposition 5.4 (Contested compliance of $\mathcal{K}_{taw}$). *Under the organised form of Definition 5.3, $\mathcal{K}_{taw}$ is impermissible under International Islamic Fiqh Academy (2009) and AAOIFI Shariah Standard No. 30 §5. Under a non-coordinated form, in which the client independently sources the third-party buyer and the sequence is not pre-arranged by the bank, permissibility is contested and depends on the ijtihād followed. Where the markup μ is calibrated against an interbank benchmark, (C3) is additionally violated mechanically by Proposition 4.1.*

This paper does not endorse organised *tawarruq*. It is recorded here for completeness of the survey of alternative architectures, and to enable the reader to place the third architecture, introduced in Section 5.4, in context.

5.4. Architecture C: Sharīkat al-ʿinān pooled-equity market access

This subsection introduces and analyses the paper’s principal constructive contribution: a market-access architecture built on the classical Islamic contractual joint-equity partnership (*sharīkat al-ʿinān*, a species of *sharīkat al-ʿaqd*) that is formally consistent with the operational design constraints of Definition 5.1 while providing the retail client with proportional economic exposure to a chosen underlying S .

Remark 5.4 (Categorical classification: *sharīkat al-ʿaqd*, not *sharīkat al-milk*). Classical *fiqh* distinguishes two categories of partnership: *sharīkat al-ʿaqd* (contractual partnership, formed by mutual undertaking to pool capital and share profits) and *sharīkat al-milk* (proprietary co-ownership of a specific asset, which may arise incidentally, e.g. through joint inheritance). AAOIFI Shariah Standard No. 12 is expressly framed around *sharīkat al-ʿaqd* and its provisions on profit-and-loss-sharing, management, and dissolution presuppose that categorisation. The construction below is deliberately a *sharīkat al-ʿaqd*: a distinct legal entity (the partnership) formed by contractual undertaking between client and broker, into which capital is contributed, which then acquires and holds the venture’s assets through a segregated custody arrangement. The parties’ interests are proportional beneficial-and-profit shares in the partnership itself, not direct individual co-ownership shares in the underlying asset. This distinction matters both for the applicability of AAOIFI SS No. 12 and for the qabḍ analysis in Proposition 5.6.

Definition 5.4 (*Sharīkat al-ʿinān* market-access architecture). The *sharīkat al-ʿinān* market-access architecture $\mathcal{K}_{\text{shr}}$ on underlying S comprises the following. The construction is a *sharīkat al-ʿaqd* (Remark 5.4) supported by an independent management-services agreement (Remark 5.5).

At contract inception t_0 :

1. Client contributes capital $K_c > 0$ and broker contributes capital $K_b > 0$ to a contractual partnership (*sharīkat al-ʿinān*, a species of *sharīkat al-ʿaqd*) formed by executed partnership deed. The strict positivity of both contributions is what makes the structure a genuine *sharīkah*; the limiting case $K_b = 0$ is treated separately in Remark 5.6.

2. Using the pooled capital $K_c + K_b$, the partnership acquires N units of the underlying at spot price S_{t_0} , where

$$N = \frac{K_c + K_b}{S_{t_0}}. \quad (32)$$

3. The N units are held by a segregated nominee/custodian acting on behalf of the partnership under an arrangement that satisfies the operational constructive-possession criteria of Proposition 5.6. Each partner holds a proportional beneficial-and-profit-share interest in the partnership itself (fractions $K_c/(K_c + K_b)$ and $K_b/(K_c + K_b)$) rather than a direct individual co-ownership share of the underlying. Realising this in practice requires: (i) a custody arrangement that safeguards partnership assets separately from the broker's proprietary balance sheet, under the applicable domestic safe-custody regime (in Australia, the ASIC provisions governing custodial services, including RG 133 on funds management and custodial services; in the United Kingdom, the FCA safe-custody-assets regime, principally CASS 6), (ii) a bilateral partnership deed under the governing law that formally identifies the shares as *sharīkat al-'inān* interests in the partnership, and (iii) alignment between the accounting treatment on the broker's books and the partnership's substantive economic reality. Whether any particular jurisdictional implementation satisfies all three is a matter of legal analysis outside the paper's scope; see Section 7.4.
4. Profit-share ratio (π_c, π_b) is pre-agreed with $\pi_c + \pi_b = 1$; per AAOIFI Shariah Standard No. 12, *losses* must be allocated strictly in proportion to capital contribution,

$$\ell_c = \frac{K_c}{K_c + K_b}, \quad \ell_b = \frac{K_b}{K_c + K_b}, \quad (33)$$

while *profits* may be distributed under the agreed permissible profit-sharing ratio (π_c, π_b) , which need not equal the capital ratio.

5. Broker acts as *managing partner* of the *sharīkah*. Any compensation for broker management services is contracted for and paid under a separate, independent management-services agreement (see Remark 5.5), not as internal remuneration under the partnership deed itself. Under that agreement the broker acts as *wakīl* for a defined *ujrah* rate $u \geq 0$ per unit time per unit of client capital, with no functional dependence on

interbank benchmark rates. The cumulative *ujrah* payable by the client over the holding period $[t_0, t_1]$ of length $\tau = t_1 - t_0$ (in years) is

$$u_c^{\text{shr}} := K_c \cdot u \cdot \tau. \quad (34)$$

At exit time t_1 :

6. Position is liquidated at spot price S_{t_1} ; the total proceeds $N \cdot S_{t_1}$ are distributed to the parties in accordance with the pre-agreed profit-and-loss rules (steps 4–5).

The eligible-underlying set is any asset that admits transfer to the partnership's segregated custody at spot: exchange-listed equities (subject to AAOIFI Standard No. 21 sectoral and financial-ratio screening), spot commodities held via allocated account or physical custody, and spot FX delivered to segregated bank accounts. Cash-settled indices, non-deliverable forwards, volatility instruments, and futures contracts (which are not themselves the underlying) fall outside the eligible set; a post-physical-delivery construction on deliverable-underlying futures is a distinct architecture that is not analysed here.

Remark 5.5 (Two-contract construction: partnership plus independent management-services agreement). AAOIFI Shariah Standard No. 12 does not permit fixed remuneration for management to be embedded directly in a *sharīkah* partnership contract; it does, however, permit a partner to provide management services under a *separately executed* contract of agency (*wakāla bi'l-ujrah*) or *ijāra*, under which remuneration is payable. Definition 5.4 adopts this two-contract construction:

- (a) a *partnership deed* that documents capital contributions, profit-and-loss shares, decision rights, and dissolution;
- (b) a *separate management-services agreement* executed alongside the partnership, under which the broker (in the role of *wakīl*) provides defined execution, custody-coordination, and administration services to the partnership for the *ujrah* u_c^{shr} of (34).

The management-services agreement must specify at least: scope of services, fee schedule, agency duties owed to the partnership, termination conditions, treatment of conflicts of interest, liability for breach, whether the fee is payable independent of venture profit, and an express undertaking not to guarantee

partnership capital. Adopting this two-contract construction is what allows the *ujrah* to be legitimate compensation for a separately identifiable service (as required by SS No. 12) rather than a disguised guaranteed return on the broker's partnership capital. It also addresses the concern that the broker's aggregate compensation combines partner returns and manager fees: the two flows arise from formally distinct contracts.

Remark 5.6 (Limiting case: $K_b = 0$ is not a *sharīkah*). If $K_b = 0$, the broker contributes no partnership capital, the structure of Definition 5.4 is not a *sharīkat al-'inān* but rather a *wakāla* (agency): the client owns the entire venture, and the broker acts purely as agent-manager for a fee. That arrangement is also permissible under Shariah, but under a different nominate-contract category with different governance and disclosure requirements. The propositions in this subsection are stated for the genuine $K_b > 0$ *sharīkah* case; the $K_b = 0$ limit is discussed as a distinct architectural variant.

We now derive the client's net payoff and establish the paper's formal consistency claims.

Remark 5.7 (Governance concentration: broker as managing partner, *wakīl*, and custody-coordinator). Definition 5.4 concentrates three distinct roles in a single institution: the broker is simultaneously a capital partner in the *sharīkah* (contributing $K_b > 0$), the appointed managing partner acting as *wakīl* under the separate management-services agreement of Remark 5.5 (receiving the *ujrah* u_c^{shr}), and, via its coordination of the segregated custody arrangement, an intermediary whose relationship with the custodian may allow it effective operational influence over the partnership's assets. This concentration creates well-known conflict-of-interest concerns: the broker as managing partner may have discretion over trading, hedging, or liquidation decisions whose outcomes affect its own partner return, and its coordination role on custody exposes the partnership to operational risk if the custody arrangement is not properly independent. Meaningful mitigation of these conflicts requires at minimum (a) written governance-rights allocation in the partnership deed (voting rights on material decisions; scope of managing-partner discretion; grounds for removal), (b) independent Shariah Supervisory Board (SSB) oversight of managing-partner conduct per AAOIFI Governance Standard No. 1 (Accounting and Auditing Organization for Islamic Financial Institutions, 2015a), (c) genuine separation of the custodian from the broker's balance sheet under the applicable safe-custody regime, and (d) transparent

disclosure to the client of the broker's combined economic exposure across partnership and management-services agreements. Whether the resulting governance package adequately discharges the fiduciary duties expected of a managing partner and *wakīl* in classical *fiqh* is a matter for the SSB reviewing the specific implementation, not for this paper.

Proposition 5.5 (*Sharīkat al-'inān* payoff dynamics). *Under Definition 5.4 with proportional profit-and-loss sharing $\pi_c = \ell_c = K_c/(K_c + K_b)$, the client's net payoff is*

$$\Pi_c^{shr}(t_0, t_1) = K_c \cdot \left(\frac{S_{t_1}}{S_{t_0}} - 1 \right) - u_c^{shr}. \quad (35)$$

Proof. The joint venture's gross P&L at t_1 is $N \cdot (S_{t_1} - S_{t_0})$. Under proportional sharing, the client receives fraction $K_c/(K_c + K_b)$ of this quantity. Substituting for N from (32) gives

$$\pi_c \cdot N \cdot (S_{t_1} - S_{t_0}) = \frac{K_c}{K_c + K_b} \cdot \frac{K_c + K_b}{S_{t_0}} \cdot (S_{t_1} - S_{t_0}) = \frac{K_c}{S_{t_0}} \cdot (S_{t_1} - S_{t_0}) = K_c \cdot \left(\frac{S_{t_1}}{S_{t_0}} - 1 \right).$$

Subtracting the broker's *ujrah* gives (35). □ □

Remark 5.8 (Economic equivalence to a spot purchase, at the gross level). Proposition 5.5 establishes that, under proportional profit-and-loss sharing, the client's *gross* exposure to the underlying is exactly that of a direct spot purchase of K_c/S_{t_0} units of S . The *ujrah* $u_c^{shr} = K_c u \tau$ is the incremental cost of the partnership structure and is bounded below by zero. At the *gross venture* level, the client and broker are on the same side of the position and gross P&L is shared in proportion to capital contribution. At the *net individual* level after *ujrah*, small positive venture returns can leave the client with negative net P&L while the broker receives the *ujrah* on top of its share of the venture return; the gross-level partnership alignment is what matters for the *maysir* analysis, and the fee is separately treated as legitimate compensation for the agency service. This is a substantively different situation from the CFD case (Proposition 3.3), where the client's gain is the broker's loss by construction and there is no shared venture.

Remark 5.9 (No leverage *within this architecture*; lower bound on net payoff). Definition 5.4 does not admit leverage: the client's effective exposure to the underlying is bounded above by $K_c \cdot S_{t_1}/S_{t_0}$, and the client's net payoff Π_c^{shr} is bounded below by $-K_c - u_c^{shr}$ (attained in the limit $S_{t_1} \rightarrow 0$; the

underlying investment loss alone is bounded below by $-K_c$ and the accrued ujrah adds to that). Introducing leverage into this specific joint-ownership architecture would require either (a) debt financing of the leveraged fraction (a *qard* with *ziyādah* concern), (b) non-ownership of the leveraged fraction (a *qabd* concern), or (c) a deferred contingent obligation structure (a *bay' al-kālī' bi'l-kālī'* concern). Leverage is therefore structurally unavailable within Definition 5.4 *as constructed*; this is not a general theorem that no Shariah-compliant financing anywhere can produce leverage, but a property of this specific joint-ownership architecture under the paper's adopted constraints. Section 5.6 addresses the CFD use cases for which this absence of leverage is a binding constraint.

Proposition 5.6 (Formal consistency of $\mathcal{K}_{\text{shr}}$ with the operational design constraints). *Under Definition 5.4 with the loss-sharing constraint $\ell_c = K_c/(K_c + K_b)$ of clause 4, $K_b > 0$, u $\mathcal{G}_s^K$ -measurable per (D1) below, the two-contract construction of Remark 5.5, and provided the segregated custody arrangement of clause 3 satisfies the operational constructive-possession criteria stated in the (D2) step below, $\mathcal{K}_{\text{shr}}$ satisfies each of the operational design constraints (D1)–(D5) of Definition 5.1. Whether this formal consistency amounts to Shariah compliance in general is a question outside the scope of the proposition and belongs to a qualified Shariah supervisory board.*

Proof. We verify each design constraint in turn.

(D1) *No interest-linked consideration.* The only periodic consideration in $\mathcal{K}_{\text{shr}}$ is the ujrah rate u , which by hypothesis is measurable with respect to the non-rate σ -algebra $\mathcal{G}_s^K$ of Definition 5.1, i.e. no functional dependence on any of $r_s^{\text{CCY1}}, r_s^{\text{CCY2}}, r_s^{\text{IBOR}}$.

(D2) *Constructive possession where a sale is present.* This step is the *qabd* analysis. By clause 3 of Definition 5.4, the partnership acquires the N units of S at t_0 and holds them via a segregated nominee/custodian on behalf of the partnership. Ownership does not by itself entail *qabd* in the sense of AAOIFI Shariah Standard No. 18; constructive possession additionally requires legally recognised control and entitlement over the asset. We therefore state constructive possession as a *conditional* conclusion contingent on the custody arrangement satisfying the following operational criteria: (i) the assets are held in a segregated custody account whose title identifies the partnership (not the broker's proprietary balance sheet), (ii) the partnership has an enforceable entitlement to direct the disposal of the assets on demand, (iii) the assets are ring-fenced from the broker's creditors in the event of broker

insolvency (as required by the applicable safe-custody regime), and (iv) the nominee owes fiduciary duties to the partnership rather than to the broker in its proprietary capacity. Under any custody implementation satisfying (i)–(iv), the partnership-level control and custody conditions required for constructive possession at the level of the *partnership* are met at t_0 . Whether that partnership-level *qabd* is sufficient to establish the requisite possession for each partner’s partnership interest, in the sense demanded by AAOIFI Standard No. 18 for the relevant transaction under the particular governing law, is a distinct fiqh-and-legal question that this paper does not resolve; it is left to SSB and legal review of any specific implementation (Section 7.4). Whether a given jurisdictional implementation genuinely delivers (i)–(iv) is similarly a matter of legal analysis.

(D3) *Real-economic-activity linkage*. The joint venture’s payoff derives from the acquisition of N units of the underlying asset S at t_0 and its subsequent disposal at t_1 ; the payoff is not a cash-settled contingent claim on a price difference alone.

(D4) *Proportional risk sharing*. By clause 4 of Definition 5.4 and the proportional condition $\pi_c = \ell_c$, gross venture P&L is shared strictly in proportion to capital contribution. At the gross level, both counterparties are on the same side of the underlying position (gains and losses accrue jointly); the *ujrah* u_c^{shr} is a separately identifiable agency fee, so at the net individual level after fees the parties’ P&Ls need not have the same sign, but the gross-level partnership structure is what the *maysir* concern addresses (see Remark 5.8).

(D5) *Immediate substantive consideration*. At t_0 , both K_c and K_b are contributed in cash to the joint venture; both contributions are non-contingent, immediate, and irreversible absent liquidation. Substantive consideration is therefore rendered by both parties at inception.

Parameter constraint on gharar (C5). Under Definition 5.4, no leverage is introduced (see Remark 5.9) and consequently there is no price-triggered maintenance-margin liquidation of the kind analysed in Proposition 3.2 (contractual liquidation, redemption, or fund termination may still occur under separately specified conditions). The underlying is a deliverable real asset with observable spot pricing, eliminating the specific execution-ambiguity concern of Definition 4.2 under a standard exchange-traded custody arrangement. The terminal exposure of the underlying investment is bounded below by $-K_c$, so the client’s net payoff is bounded below by $-K_c - u_c^{\text{shr}}$.

Combining the five verifications gives the stated formal consistency. $\square$

□

Restriction on eligible underlyings.. Definition 5.4 requires that N units of the underlying S be actually deliverable to the partnership’s segregated custody. This restricts the eligible underlying set to instruments admitting such transfer:

- Spot equities via nominee custody (with additional Shariah-compliance screening on the issuer under AAOIFI Standard No. 21);
- Spot commodities via warrant, allocated account, or physical custody;
- Spot FX via delivery to segregated bank accounts.

It excludes: cash-settled indices (S&P 500, DAX, VIX); non-deliverable forwards on restricted currencies; futures contracts (which are not themselves the underlying deliverable asset); and synthetic instruments such as variance swaps, dispersion products, and volatility indices. A distinct post-physical-delivery construction on deliverable-underlying futures is imaginable but is a separate architecture not analysed here. Section 5.6 addresses the resulting design gap.

5.5. Comparative architecture map

Table 2 summarises the three alternative architectures against a common set of dimensions.

The comparison exposes two structural features common to all three candidate architectures: (i) none admits leverage in excess of $1\times$ on the client’s capital, and (ii) all require deliverable, real underlyings and are therefore incompatible with purely synthetic instruments. The economic cost of these features against the conventional leveraged CFD baseline is quantified in Section 6.

5.6. CFD use cases with no permissible equivalent

We conclude the section by stating honestly which retail CFD use cases have *no* permissible equivalent under the constraint set $\mathcal{C}_{\text{Shariah}}$. This is a finding, not a limitation of the paper: the incompatibility is structural, and identifying it precisely is itself a research contribution.

The following use cases have no permissible substitute in Definition 5.1:

Table 2: Comparative properties of the three alternative architectures. “Effective leverage” refers to notional exposure per unit of client capital; “scholarly status” summarises the consensus recorded in Sections 5.2–5.4.

Architecture	Use case served	Eligible underlying	un-	Effective leverage	Scholarly status
$\mathcal{K}_{\text{wa'd}}$	FX forward exposure; commercial hedging	Deliverable FX pairs		$1\times$	Contested (BNM permits, GCC/AAOIFI restrictive)
$\mathcal{K}_{\text{taw}}$	Cash financing synthesis	LME-listed commodities		N/A (financing, not exposure)	Impermissible in organised form (OIC 179); contested otherwise
$\mathcal{K}_{\text{shr}}$	Spot directional exposure; buy-and-hold	Deliverable equities, spot FX, spot commodities		$1\times$	Formally consistent with (D1)–(D5) (Prop. 5.6); subject to SSB review

1. *Leveraged short-horizon directional speculation.* Any use case requiring effective exposure in excess of $1\times$ client capital is precluded by Remark 5.9: leverage is structurally incompatible with the joint constraint set (C3), (C6), (C7).
2. *Cash-settled synthetic index exposure.* Any use case requiring exposure to a non-deliverable underlying (index, volatility measure, variance swap) is precluded by Definition 5.4 clauses 2–3, which require an actually-deliverable and jointly-ownable underlying.
3. *Short-selling of non-owned assets.* The classical short sale, selling an asset one does not own with the intention of buying it back later at a lower price, violates Standard No. 18 (*qabḍ*) and cannot be reconstructed under the joint-equity framework.
4. *Pure intraday price-difference wagers on unbacked references.* Retail contracts that transact on a reference price with no underlying commercial rationale and no contemplation of delivery, often marketed as “binary options” or “exotic CFDs”, lie outside the domain of any of the three architectures.

The paper’s position on these use cases is that their unavailability under

the joint-ownership architecture of Definition 5.4 is a feature of that specific construction rather than a limitation. The retail-outcome evidence surveyed in Section 1.1, retail CFD loss rates of 74–89 per cent across major jurisdictions, is associated with substantial realised losses among retail participants in these product categories. Reasonable readers may reach different normative conclusions about the wider policy implications; the paper records the empirical evidence and its convergence with the leverage-cap feature of the proposed architecture without extrapolating to broader normative claims.

Section 6 quantifies the economic cost of the candidate architectures, principally $\mathcal{K}_{\text{shr}}$, against a conventional leveraged CFD baseline, under illustrative Bates parameterisations.

6. Quantitative Comparison via Monte Carlo Simulation

Sections 3–5 established the theoretical basis for the paper’s central claim: the standard retail CFD exhibits five structural concerns under the adopted AAOIFI/OIC constraint framework, and a *sharīkat al-’inān* pooled-equity architecture is formally consistent with the operational design constraints (D1)–(D5) of Definition 5.1 subject to SSB review, at the cost of eliminating leverage. This section quantifies the economic cost of that trade-off under a common jump-diffusion price process, using a Monte Carlo simulation across four underlyings, four holding horizons, and four leverage regimes.

Framing note. The comparison in this section is *not* a claim that the *sharīkat* architecture is intrinsically superior to a leveraged CFD as a financial product: the two architectures deliver different economic exposures on identical client capital and are therefore not directly comparable on a like-for-like basis. Concretely, a client with capital K_c takes exposure $K_c \cdot L/S_{t_0}$ units of the underlying under a CFD at leverage L , versus K_c/S_{t_0} units under the *sharīkat* at compulsory unit leverage. The comparison should therefore be read as a quantification of the *compliance cost* of the Shariah-imposed leverage cap: relative to a matched-capital leveraged CFD, the compliant architecture forgoes proportional leveraged upside and leveraged downside, and eliminates margin-call risk. Whether this trade-off is desirable for a given participant depends on their risk preferences and religious commitments, not on any intrinsic superiority claim. All figures and tables in this section derive from the reference implementation released with the paper (Section 6.8).

6.1. Simulation design

The simulation is organised as a $4 \times 4 \times 4$ grid over underlyings, holding horizons, and leverage regimes, with $n = 10,000$ paths per cell.

Underlyings.. Three major FX pairs (EUR/USD, USD/JPY, AUD/USD) and one equity-index proxy (S&P 500) span the volatility range typical of retail CFD flow. Each is modelled under the Bates (1996) jump-diffusion of Assumption 3.1; illustrative parameter calibrations, consistent with published FX and equity implied-volatility surfaces (Andersen, 2008; Bates, 1996), are recorded in Table 3.

Table 3: Bates model calibration presets used in the Section 6 simulations. Long-run volatility $\sigma_{\text{LR}} = \sqrt{\theta}$; the initial variance is set to $v_0 = \theta$ (spot start at long-run equilibrium); the drift is set to $\mu = 0$ for all underlyings to isolate the compliance cost from directional views; the mean-reversion speed is $\kappa_v = 2.0 \text{ yr}^{-1}$ and the vol-of-vol is $\sigma_v = 0.3\sqrt{\theta}$ for all underlyings, chosen to satisfy the strict Feller condition $2\kappa_v\theta > \sigma_v^2$. FX interbank overnight rates r^{CCY1} , r^{CCY2} used in the CFD Tom-Next computations are the illustrative fixed values SOFR = 5.25%, ESTR = 3.50%, SONIA = 5.00%, TONA = 0.10%, RBA cash rate = 4.35%, held constant across the simulation horizon. The S&P 500 case in Section 6.5 is included as a mathematical counterfactual for isolating the compliance-cost differential across asset-class volatility regimes: a Shariah-compliant equity implementation would use a Standard No. 21-screened basket or ETF rather than the index itself. Complete numerical values are also machine-readable in `src/bates.py` of the reference implementation.

Underlying	σ_{LR}	ρ	λ (jumps/yr)	μ_J	σ_J
EUR/USD	8%	−0.10	1.5	−0.002	0.012
USD/JPY	9%	+0.05	2.0	+0.001	0.015
AUD/USD	11%	−0.20	2.5	−0.003	0.018
S&P 500 [†]	18%	−0.65	3.0	−0.015	0.035

Holding horizons.. Four horizons span the retail trading spectrum, expressed in trading days: 1 day (intraday exit), 5 days (one trading week), 21 days (approximately one trading month), and 63 days (one trading quarter). All time-in-year conversions in Section 6 use the trading-day convention $\tau_{\text{sim}}(h) = h/252$; this differs from the calendar/actual-365 convention $\tau = 1/365$ used in the Tom-Next derivation of Section 3.3, which follows industry practice for overnight financing. The simulation thus mixes conventions across two logically distinct operations: the Bates path generator and the *sharīkat* ujah accrual use trading-day-per-year, while the CFD Tom-Next fee accrual uses

calendar-day-per-year. Both are correctly implemented in the reference code (see `src/simulate.py`) and the mixing has been checked to leave the Section 6 comparisons unaffected in direction, though absolute per-day figures differ between the two operations by the ratio $252/365 \approx 0.69$; the *sharīkat* 63-day ujah figure of approximately USD 25 in Figure 1 follows from $K_c \cdot u \cdot \tau_{\text{sim}}(63) = 10,000 \cdot 0.01 \cdot (63/252) = 25$.

Leverage regimes.. Four regimes are considered: $L \in \{1, 5, 10, 30\}$. The maximum $L = 30$ corresponds to the ASIC 2020 product-intervention cap on retail *major-FX* CFDs specifically (Australian Securities and Investments Commission, 2020); the same $30\times$ ceiling applies to retail major-FX CFDs under the analogous ESMA (European Securities and Markets Authority, 2018) and FCA (Financial Conduct Authority, 2019) product intervention frameworks. Lower caps apply to other retail CFD categories under all three regimes (typically $20\times$ for non-major FX and gold, $10\times$ for other commodities and major equity indices, $5\times$ for individual equities, and $2\times$ for cryptocurrencies). The $L = 30$ regime in this simulation should therefore be read as the ceiling that applies to the FX subset of the underlyings studied here; the same regime applied to the S&P 500 exceeds what any of the three regulators permit for retail index CFDs, and is included in the grid as a stress case rather than as a representation of permitted retail exposure.

Contract parameters.. Client capital is fixed at $K_c = \text{USD } 10,000$. The CFD baseline uses a maintenance margin coefficient $\alpha = 0.5$, and stylised common baseline transaction-cost assumptions of a 2 bps bid-ask spread per side and a 100 bps per annum broker markup on the interbank differential; these are illustrative values applied uniformly across the four underlyings for comparability and are not intended as representative universal retail-CFD pricing (FX, equity indices, and single-name equities have materially different spread and funding structures in practice). The *sharīkat al-'inān* architecture uses a client-side ujah of 100 bps per annum, flat per unit of holding period, independent of interbank rates. Interbank rate presets (r^{CCY1} , r^{CCY2}) are set to representative 2025 levels for each pair.

Numerical scheme.. The Bates system is discretised by Euler–Maruyama on the log-price with full-truncation of the variance process to preserve positivity, following Andersen (2008). All path simulations use fixed seeds derived from a session-wide base seed, so the reported results are exactly reproducible via the released code.

6.2. Metrics

For each grid cell we compute:

- *All-in holding cost*: the sum of expected financing and per-entry-and-exit transaction costs (spread plus commission) over the holding period, in USD. The naming distinguishes this quantity from cost-of-carry in its strict sense, since the spread component accrues at entry and exit rather than continuously.
- *Margin-call trigger probability*: the empirical frequency $\mathbb{P}(\tau_{MC} < t_1)$ estimated as the fraction of paths that cross the barrier of (16) prior to the horizon.
- *Return-distribution moments*: mean, standard deviation, skewness, and excess kurtosis of the net client P&L.
- *Tail-risk measures*: Value-at-Risk and Conditional Value-at-Risk at the 95% level; per-path maximum drawdown.
- *Risk-adjusted performance*: annualised Sharpe and Sortino ratios.
- *Loss frequency*: the empirical probability of a negative net client P&L over the holding period.

Note on the equity-index case. The Tom-Next financing formula of Section 3.3.1 is derived under FX covered interest rate parity, with r^{CCY1} and r^{CCY2} interpreted as base- and quote-currency overnight rates. For the S&P 500 case in the simulation grid, the natural financing structure is instead $r^{\text{funding}} - q^{\text{dividend}}$, where r^{funding} is the client's funding rate on the leveraged position and q^{dividend} is the index dividend yield. In the simulation we take $r^{\text{funding}} \approx r^{\text{USD}}$ and $q^{\text{dividend}} = 0$ as an illustrative simplification solely to isolate the pure leverage effect from dividend-carry effects; a realistic S&P 500 dividend yield is of order 1.3%, so a full implementation would apply $r^{\text{funding}} - q^{\text{dividend}}$ with that value. The simplification does not affect the direction of the compliance-cost comparison but does reduce the absolute CFD financing cost on the equity leg by that fraction. These are not empirical calibrations and produce financing figures broadly comparable in magnitude to a USD-quoted FX pair, which is sufficient for the compliance-cost comparison that is the section's object. A production implementation would require an equity-specific financing derivation, deferred to subsequent work.

6.3. Statistical inference

Percentile bootstrap 95% confidence intervals are computed for all mean-P&L estimates using 500 bootstrap resamples per cell. For the binary margin-call-probability estimate at each cell, Monte Carlo standard error is approximately $\sqrt{\hat{p}(1 - \hat{p})/n}$ with $n = 10,000$; a representative point estimate $\hat{p} = 0.48$ carries a standard error of approximately 0.005 (i.e. half a percentage point). Bootstrap confidence intervals on the tail metrics VaR_{95} and CVaR_{95} are computed in the code but are not tabulated here for brevity; they are of order $\pm 3\%$ – 5% of the point estimates and do not affect the qualitative conclusions. The Kolmogorov–Smirnov two-sample statistic is reported as a simulation-based distributional-distance measure between CFD and *sharikat* return distributions; a small KS value indicates limited simulation-based distributional separation but is not, and is not interpreted as, formal evidence of distributional equality. p -values are adjusted for multiple comparisons using the Benjamini–Hochberg false-discovery-rate procedure at $q = 0.05$ (Benjamini and Hochberg, 1995).

6.4. Machine learning extension: what drives margin-call events?

To characterise the price-path features that drive CFD tail losses we train an XGBoost gradient-boosted classifier (Chen and Guestrin, 2016) on path-level features to predict whether a given path will trigger a margin call over a 21-day horizon at $L = 10$ on the S&P 500 underlying. Features are constructed from early-path information, first-day log return, absolute log return, and realised variance; 3-day log return and mean variance, so the classifier answers the operationally relevant question of whether a margin call is predictable from early observations of the path.

The classifier achieves out-of-sample area under the ROC curve of $\text{AUC} = 0.728$ on a 25% test partition (class balance 27.7% margin-called). Figure 5 displays the gain-based feature importances; the 3-day cumulative log return carries the largest gain in the model, followed by 3-day mean variance, with first-day features contributing marginally. This is model-consistency evidence: within the stylised simulated environment, a simulated margin-call event is partially predictable from simulated early returns and realised variance, and the classifier’s largest gains come from features aggregating information over several days rather than any single-day realisation. The result does not identify a live-market empirical relationship; the exercise demonstrates conditional predictability of a simulated barrier-crossing event from simulated early-path information, consistent with the barrier structure of Proposition 3.2.

6.5. Results

Cost of carry.. Figure 1 shows the expected cost of carry for EUR/USD over the four horizons at leverages $L \in \{5, 10, 30\}$ for the CFD, alongside the flat ujrah cost of the *sharikat* architecture. The CFD's cost scales approximately linearly in leverage and in horizon; at $L = 30$ and a 63-day horizon it reaches USD 1,220, compared to USD 25 for the *sharikat* at the same horizon (a factor of approximately $49\times$).

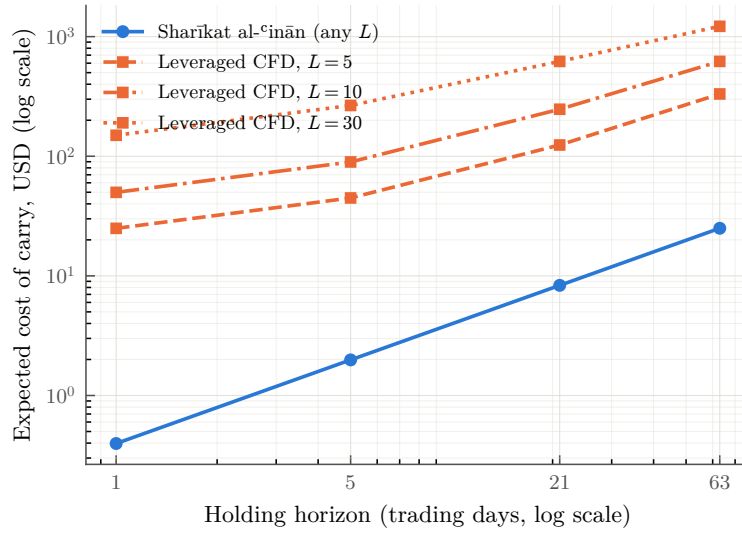


Figure 1: Expected cost of carry as a function of holding horizon for the EUR/USD underlying under the standard retail CFD at three leverage regimes and under the *sharikat al-'inān* architecture (which is leverage-invariant). Both axes are logarithmic. The *sharikat* cost line is unique, whereas the CFD produces one line per leverage regime. The 100-bps annual ujrah accrues linearly with horizon; the CFD cost accrues linearly with the leveraged notional, giving the two lines' different elevations.

Margin-call frequency.. Figure 2 presents the margin-call trigger probability $\mathbb{P}(\tau_{MC} < t_1)$ across the full grid. Three qualitative features emerge: (i) at $L = 1$, no margin calls were observed in any of the 10,000 simulated paths (the barrier lies approximately 50% below the entry price and, under the parameters of Table 3, an excursion of that magnitude is extremely unlikely within the horizons studied, though not mathematically impossible); (ii) at $L = 30$ and horizon 63 days, margin-call frequency exceeds 60% for every underlying and reaches 74.5% for the S&P 500 (the highest-volatility underlying); (iii) the S&P 500 dominates every other underlying at every

(L, h) combination beyond the first row/column, driven by its higher long-run volatility and stronger leverage effect ρ .

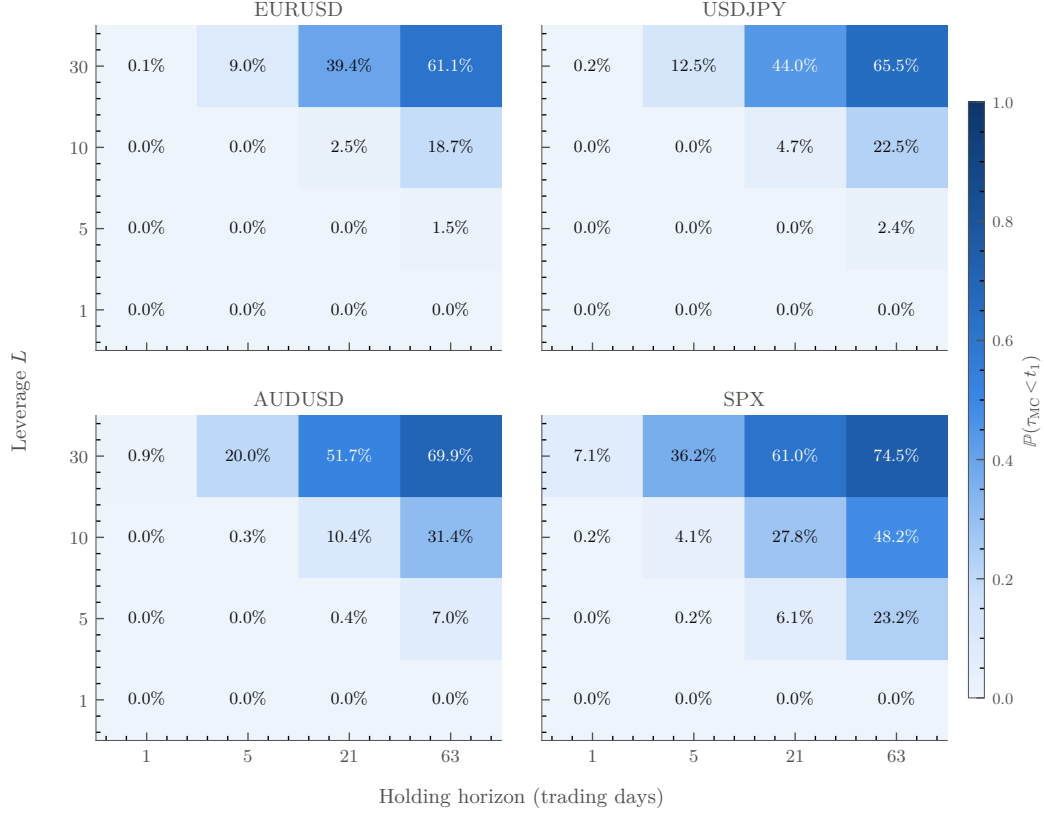


Figure 2: Margin-call trigger probability $\mathbb{P}(\tau_{MC} < t_1)$ estimated over $n = 10,000$ Monte Carlo paths per grid cell, across the four underlyings of Table 3. Cell shading indicates the estimated probability; numerical annotations give the point estimate.

Return-distribution comparison.. Figure 3 contrasts the empirical CDFs of net client P&L for the AUD/USD underlying at a 21-day horizon. The *sharīkat* CDF is narrowly concentrated around zero, reflecting the unleveraged exposure to spot AUD/USD returns over one month. The leveraged CFD CDF at $L = 10$ is markedly wider, with pronounced left-tail mass driven by margin-call events. Empirical 95% VaR statistics from the full grid, reported in Table 4, formalise this visual comparison.

Tail-risk decomposition: jumps vs. diffusion.. To quantify the contribution of the jump component to CFD tail losses, Figure 4 compares the left tail

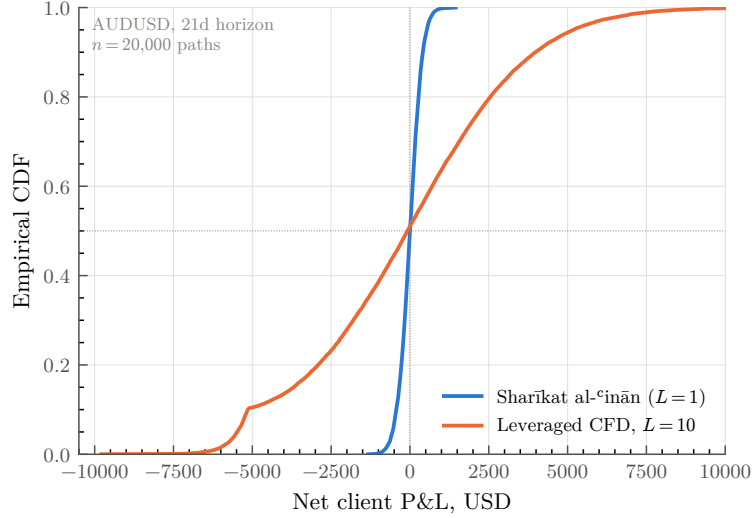


Figure 3: Empirical cumulative distribution functions of net client P&L: leveraged CFD (orange) at $L = 10$ versus the *sharikat al-'inān* architecture (blue) at $L = 1$ on AUD/USD over a 21-day horizon. Sample size $n = 20,000$ paths per architecture.

(below the 20th percentile) of the CFD P&L distribution on S&P 500 at $L = 10$ under the full Bates model against the same simulation with the jump intensity set to zero ($\lambda = 0$). The full-model tail exhibits a pronounced spike at the negative-balance-protection floor (loss of USD 10,000, the initial margin), which the diffusion-only counterpart does not produce.

Machine-learning feature importances..

Tabular summary.. Table 4 consolidates the tail-risk metrics across the four underlyings at the 63-day horizon. The *sharikat* 95% VaR does not vary with the CFD leverage parameter L (the *sharikat* has no leverage parameter), and ranges from USD 687 (EUR/USD) to USD 1,554 (S&P 500 counterfactual, see Table 3 footnote) driven by the underlying volatilities. The CFD 95% VaR grows approximately linearly in leverage and reaches the initial-margin floor at $L = 30$ for the S&P 500. Loss frequency for the CFD remains near, but need not equal, 50% at unit leverage; the observed $L = 1$ figures (approximately 45–57% across the four underlyings) reflect return skewness, jump asymmetry, financing carry and transaction costs, none of which the zero-drift condition eliminates. The frequency rises to 69–76% at $L = 30$ (dominated by margin-call events), whereas the *sharikat* loss frequency remains close to 50% across

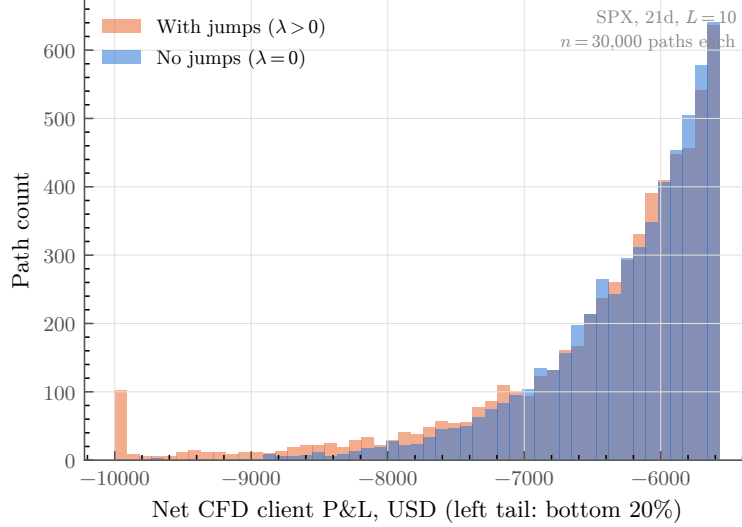


Figure 4: Left-tail distribution of net CFD client P&L on the S&P 500 underlying at $L = 10$ and 21-day horizon, comparing the full Bates model (with jumps) against the diffusion-only counterpart (no jumps). Both simulations use the same random seeds and identical Brownian innovations, differing only in the presence of the Poisson jump component. The spike at $-10,000$ USD (initial margin) under the full model reflects margin-call events that discontinuous price movements contribute to; the no-jump distribution does not produce a comparable spike over the same paths, indicating that jumps make a substantial contribution to the extreme left tail under the illustrative parameterisation used, without implying that jumps are the sole cause of extreme losses.

all leverage regimes (again, because the *sharīkat* has no leverage regime).

Kolmogorov–Smirnov distributional tests. The KS two-sample statistic between CFD and *sharīkat* return distributions ranges from 0.02–0.09 at $L = 1$ (distributions statistically indistinguishable in economic magnitude, though the very large sample sizes yield tiny p -values in most cells) to 0.47–0.75 at $L = 30$ (distributions substantially different in both location and shape, with p -values below machine precision). The magnitude of the KS statistic thus tracks the leverage regime almost monotonically: at $L = 1$ the architectures are near-equivalent in their return distribution; at $L = 30$ they are fundamentally different objects.

6.6. Interpretation: the shape of the compliance-cost curve

We reiterate the framing note at the start of this section: the CFD and the *sharīkat* deliver different economic exposures on identical client capital, so the

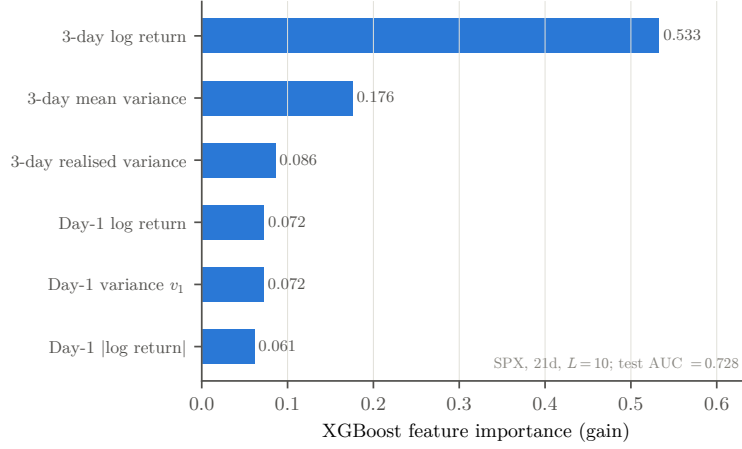


Figure 5: XGBoost feature importances for the margin-call prediction task described in Section 6.4. Test-set AUC of 0.728 on the S&P 500, 21-day horizon, $L = 10$ configuration. Under the classifier trained on this stylised simulation, the 3-day cumulative log return carries the largest gain-based feature importance, followed by 3-day mean variance. Values reflect the classifier’s use of the features within the simulated environment; they are model-specific importances, not causal or empirical structural discoveries.

results below are interpreted as characterising the shape of the compliance-cost curve, not as a superiority claim.

Result 1: at $L = 1$, the two architectures produce closely similar simulated return distributions under unit exposure. The small KS statistics at $L = 1$ (0.02–0.09 across all underlyings and horizons) indicate limited simulation-based distributional separation between the *sharīkat* at unit leverage and an unlevered CFD. The two contracts are structurally distinct, the CFD is cash-settled synthetic exposure, the *sharīkat* is proportional joint ownership, but at unit exposure their simulated payoff distributions are close. They differ in all-in holding cost structure (broker *ujrah* versus CFD spread plus financing). At this leverage regime the compliance cost is minimal.

Result 2: the cost-of-carry crossover at approximately $L \geq 5$ on EUR/USD at any horizon above one week. Because the CFD’s spread and financing scale with the leveraged notional while the *sharīkat* *ujrah* does not, the CFD’s expected carrying cost overtakes the *sharīkat*’s at moderate leverage. This is a genuine efficiency observation about the fee structure: even a client indifferent to religious constraint would find the CFD costly to hold across settlement cutoffs at $L \geq 5$ over multi-day horizons, because the spread–financing combination compounds. Note, however, that the client is receiving

Table 4: Tail-risk and loss-frequency metrics at the 63-day holding horizon across the four underlyings. VaR reported in USD (client capital $K_c = 10,000$). *Sharikat* figures are shown once per underlying since they are leverage-invariant. Simulation size: $n = 10,000$ paths per cell. [†]The *sharikat* S&P 500 row is a mathematical counterfactual benchmark used to isolate the effect of the underlying’s volatility regime; cash-settled indices are outside the eligible-underlying set of Definition 5.4, so this row does not represent an eligible implementation, which would require a Shariah-screened deliverable equity basket or equivalent eligible underlying.

Underlying	Architecture	L	$\mathbb{P}(\tau_{MC} < t_1)$	VaR _{95%}	P(loss)	CVaR _{95%}
EUR/USD	<i>sharikat</i>	1	–	687	0.52	917
	CFD	5	1.5%	3,689	0.57	4,589
	CFD	10	18.7%	5,882	0.57	6,217
	CFD	30	61.1%	8,080	0.69	8,867
USD/JPY	<i>sharikat</i>	1	–	742	0.52	995
	CFD	5	2.4%	3,379	0.45	4,402
	CFD	10	22.5%	5,201	0.46	5,529
	CFD	30	65.5%	7,182	0.67	8,052
AUD/USD	<i>sharikat</i>	1	–	951	0.52	1,252
	CFD	5	7.0%	5,255	0.53	5,492
	CFD	10	31.4%	6,130	0.55	6,630
	CFD	30	69.9%	8,812	0.74	9,589
S&P 500 [†]	<i>sharikat</i>	1	–	1,554	0.43	2,140
	CFD	5	23.2%	6,107	0.49	6,582
	CFD	10	48.2%	7,578	0.58	8,479
	CFD	30	74.5%	10,000	0.76	10,000

$L \times$ the exposure for that fee, so the per-unit-of-exposure cost comparison is more nuanced than the total-cost line suggests.

Result 3: exposure-driven tail-risk differential at $L \geq 10$. At $L = 10$ on the S&P 500 over 63 days, the CFD has a 48% probability of hitting the margin-call barrier and a 95% VaR of USD 7,578, versus a *sharikat* VaR of USD 1,554. The approximately five-fold differential in tail risk reflects two compounding effects: the CFD delivers $10 \times$ the notional exposure, and it carries a maintenance-margin close-out mechanism that the unleveraged *sharikat* structurally does not. The first is a direct consequence of the different exposures; the second is architecture-specific and represents a genuine risk feature (not merely a scaling of exposure). Note that the absence of a margin-

call mechanism in the *sharīkat* architecture does not mean the client is exposed to no forced-sale risk of any kind: the partnership deed will typically specify predefined exit or wind-down conditions (loss thresholds, term limits, mutual-consent liquidations), and pooled physical assets may need to be sold to satisfy withdrawal requests. What is absent is the leverage-driven, price-triggered close-out that occurs when maintenance-margin buffer is exhausted, which is a specific mechanism proper to leveraged margin trading and is the source of the tail-risk asymmetry documented in Section 6.5. At $L = 30$ the CFD saturates the negative-balance protection floor with probability 75%, producing a large point mass in the return distribution at the model-imposed maximum-loss threshold.

The three results together characterise the compliance-cost trade under the Shariah leverage cap: a Muslim participant who chooses the compliant architecture over a leveraged CFD accepts proportionally reduced exposure (both upside and downside), pays a flat *ujrah* in place of leverage-scaled financing, and eliminates the maintenance-margin close-out mechanism. Whether this trade is desirable for a given participant depends on their risk preferences, capital, and religious commitments. What the results *do* demonstrate is that the trade is quantitatively well-defined and that the compliant architecture, at holding horizons of one week or longer, achieves risk-adjusted properties (lower tail risk per dollar of client capital, no price-triggered maintenance-margin liquidation) that a secular prudential-regulatory framework has independently addressed through leverage caps and negative-balance protection (Heimer and Simsek, 2019; Australian Securities and Investments Commission, 2020).

6.7. Sensitivity analysis

To verify robustness, we vary the three most influential model parameters, jump intensity λ , vol-of-vol σ_v , and the correlation ρ between the price and variance Brownian innovations, over the ranges $\lambda \in \{0, 1, 3, 5\}$ per year, $\sigma_v \in \{0.1, 0.2, 0.4\}$, and $\rho \in \{-0.8, -0.4, 0, +0.4\}$, on the reference S&P 500 configuration at $L = 10$, 21-day horizon. The qualitative pattern of Section 6.5 is preserved across the sensitivity grid: the CFD's tail-risk profile worsens monotonically in λ , mildly in σ_v , and increasingly as ρ becomes more negative (the equity-style leverage effect that amplifies downside). The *sharīkat* architecture's return distribution naturally shifts with the same parameters, it depends on S_{t_1} which depends on $(\lambda, \sigma_v, \rho)$, but its *margin-call* and *maximum-loss* properties are invariant to the leverage parameter L (which is fixed at unity by construction). This is a leverage-invariance property,

not an underlying-process-invariance property; both should be kept clearly distinguished. Full sensitivity tables are reproduced in the code repository (Section 6.8).

6.8. Code and reproducibility

A reference Python implementation supporting every figure, table, and numerical claim in this section is released with the paper under the MIT License. The repository is hosted at

<https://github.com/Muhammad-Taha-Asif/Shariah-CFE-1>

and the tagged initial release (v0.1.0) is archived at Zenodo with the persistent identifier

DOI: 10.5281/zenodo.22998850

The repository includes: (i) the Bates model simulator; (ii) the CFD and *sharīkat* contract mechanics; (iii) the metrics and statistical-inference modules; (iv) the XGBoost classifier and sensitivity-analysis code; and (v) the figure-generation code that produces the PDFs used in this paper. All random-number seeds are fixed for exact reproducibility. Dependencies are listed in the repository’s `requirements.txt`; the code runs on Python 3.11 and later with the standard scientific stack (NumPy, SciPy, pandas, scikit-learn, XGBoost, matplotlib).

7. Discussion and Limitations

This section situates the paper’s contributions against their proper limits: what the results say, what they do not say, and what an implementation of the proposed architectures would require beyond the formal analysis conducted here.

7.1. Scope of the paper: what it does and does not

The paper makes four positive claims, delivered in Sections 3–6 respectively: (i) a formal mathematical characterisation of the standard retail CFD architecture under a Bates jump-diffusion; (ii) formal propositions documenting five structural incompatibilities or concerns with the adopted AAOIFI/OIC constraint framework; (iii) construction of three candidate alternative architectures, *wa’d*-based, *tawarruq*-based, and *sharīkat al-’inān*,

with the third established as formally consistent with the operational design constraints (D1)–(D5), subject to SSB review; and (iv) a Monte Carlo simulation quantifying the compliance cost of the candidate *sharīkat* architecture against the conventional leveraged CFD baseline.

The paper also makes explicit non-claims that bound its interpretation. It is *not* a work of fiqh: no proposition here constitutes a scholarly ruling on the permissibility of any existing or proposed instrument. It is not a claim about Islamic finance more broadly: the analysis is restricted to retail CFD/FX and the alternative architectures for equivalent market-access use cases. It is not a claim about any specific commercial implementation: the compliance status of a real product depends on custody arrangements, execution mechanics, Shariah supervisory review, and jurisdictional regulation, none of which are within the paper’s scope. And it is not an empirical calibration exercise: the Bates parameter presets used in Section 6 are illustrative values consistent with published FX and equity volatility studies, not fits to any proprietary dataset.

7.2. Scholarly review as a precondition for implementation

Any commercial deployment of the *sharīkat al-’inān* architecture proposed in Section 5.4, or of the *wa’d*-based alternative of Section 5.2, requires prior review by a constituted Shariah supervisory board (SSB) drawn from qualified scholars. Reputable independent SSB providers include Amanie Advisors, Shariyah Review Bureau, Islamic Finance Council UK, and the internal SSBs of the major Islamic banking institutions. Certification under AAOIFI Shariah Standards, the reference framework adopted throughout this paper, would be the natural target, with jurisdiction-specific supplementary review where local regulators (Bank Negara Malaysia, the Securities Commission Malaysia, the DFSA in Dubai) operate their own frameworks.

The role this paper aims to play in that scholarly-review process is preparatory: by making the mathematical structure of both the target architecture and the constraint set explicit, the analysis reduces the interpretive burden on the reviewing scholars and supplies them with a documented technical foundation on which their *ijtihād* can operate. It does not, and cannot, substitute for that review.

7.3. Regulatory feasibility

Any commercial implementation of the *sharīkat al-’inān* architecture would require jurisdiction-specific legal and regulatory analysis; the sketch

below identifies the frameworks most likely to be relevant in each of several major jurisdictions, without purporting to be a legal opinion.

- In **Australia**, candidate legal forms include the Managed Investment Scheme or Individually Managed Account frameworks under Chapter 5C of the Corporations Act; product-specific structuring, an Australian Financial Services Licence with appropriate authorisations, and ASIC engagement would all be required (Australian Securities and Investments Commission, 2020).
- In the **United Kingdom**, Part 4A authorisation under the FCA Handbook would apply; the FCA has engaged with Islamic finance product categories through accommodating guidance and prior policy statements (Financial Conduct Authority, 2019).
- In the **United Arab Emirates**, the DFSA describes itself as a *Shariah-systems* regulator rather than a Shariah regulator, and requires firms conducting Islamic financial business to maintain a Shariah supervisory board and governance framework. The DFSA thus provides a dedicated framework for Islamic financial business but does not itself certify Shariah compliance; legal and regulatory classification of the proposed structure would require jurisdiction-specific analysis and DFSA engagement.
- In **Malaysia**, Bank Negara Malaysia's Shariah Advisory Council has previously ruled on *sharīkah*- and *wa'd*-based structures. Product-specific engagement with BNM and the Securities Commission Malaysia would be required.
- In the **United States**, the applicable regulatory regime depends on the product's legal form: off-exchange retail FX activity is regulated by the CFTC and NFA under the Retail Foreign Exchange Dealer framework; equity brokerage falls under SEC/FINRA supervision; investment-adviser activity under the Investment Advisers Act. Shariah-compliance representations would be treated as marketing statements subject to the anti-fraud provisions of the relevant regime.

Full regulatory analysis for any specific product is beyond the paper's scope. Implementation constraints are substantial (Section 7.4) and are not resolved by the paper's mathematical treatment.

7.4. Practical implementation barriers

The *sharīkat al-ʿaqd* structure of Definition 5.4 imposes real infrastructure requirements on any commercial implementation:

- *Custody.* The segregated custody of partnership assets (clause 3 of Definition 5.4) requires a nominee-custodian arrangement satisfying the operational-*qabḍ* criteria (i)–(iv) established in the (D2) step of Proposition 5.6, under the applicable safe-custody regime (in Australia, the ASIC provisions on custodial services including RG 133; in the United Kingdom, FCA CASS 6). Standard broker-dealer custody plumbing supports segregated custody in most jurisdictions but requires bespoke legal documentation identifying the partnership as the beneficial account holder and enforceable partnership rights against the custodian.
- *Settlement.* Settlement cycles vary by asset and jurisdiction: US equities moved to T+1 on 28 May 2024; UK, EU, and Australian equities remain predominantly on T+2 at the time of writing; spot FX settlement is pair- and convention-dependent (typically T+2 for major pairs, T+1 for USD/CAD). No exotic settlement mechanics are required for any of these. Physical commodity underlyings (allocated gold, LME warrants) require additional custody arrangements but remain tractable.
- *Liquidity provisioning.* The broker’s capital contribution K_b must be warehoused for the duration of the venture. This is economically analogous to prime-brokerage balance-sheet commitment but structured contractually as equity rather than debt, and requires the broker to hold sufficient regulatory capital to support the aggregate contribution across its client book.
- *Client onboarding.* Additional documentation (Shariah-compliance attestations, disclosure of joint ownership) increases onboarding friction relative to the click-through CFD account. This is a commercial constraint, not a technical one.

None of these barriers is fundamental. All are recognisable extensions of existing broker infrastructure and legal practice.

7.5. Model limitations

The Bates jump-diffusion adopted in Assumption 3.1 is a well-tested workhorse of quantitative finance, sufficient for the paper’s purpose, namely,

comparing architectures under a common realistic-enough price process, but not the last word on empirical price dynamics. Model features not captured include:

- *Regime switching.* Discrete shifts between low- and high-volatility regimes, well-documented in FX and equity markets (Hamilton, 1989), are absent from the single-regime Bates specification.
- *Order-book microstructure.* The paper’s execution model treats spread as an exogenous parameter; real-world execution involves adverse selection, latency, and inventory dynamics analysed in the microstructure literature (Kyle, 1985; Glosten and Milgrom, 1985).
- *Liquidity dry-up.* The Bates model assumes continuous execution at posted prices; in crisis conditions the bid-ask spread widens materially and depth vanishes.
- *Parameter non-stationarity.* The illustrative calibrations of Table 3 are held constant; real markets exhibit parameter drift that a production implementation would need to accommodate through regular recalibration.
- *Alternative jump distributions.* The lognormal-jump specification is standard but restrictive; heavier-tailed alternatives (double-exponential, variance-gamma) may better capture observed left-tail behaviour on some underlyings.

Extension of the analysis to any of these model refinements is straightforward given the released code base and is a natural subject for follow-up work.

7.6. Ethical considerations

A Shariah-compliant leveraged retail product would raise, in principle, the same investor-protection concerns as its conventional counterpart. The *sharīkat al-’inān* architecture of Definition 5.4, however, does not admit leverage within its joint-ownership construction (Remark 5.9). This is not a design choice imposed for prudential reasons; it is a property of the specific joint-ownership construction under the paper’s adopted constraints. The absence of leverage under this candidate architecture is associated with the empirical evidence in Section 1.1 that leveraged retail participation is associated with substantial realised losses among retail participants (Heimer

and Simsek, 2019; Barber et al., 2014); readers may draw their own conclusions about the wider implications.

At a broader level, the availability of Shariah-compliant financial products supports religious autonomy for observant participants. The programme of research this paper initiates aims to describe and formalise architectures consistent with the constraint set that a substantial share of the global population regards as binding.

8. Conclusion and Future Research

8.1. Summary of findings

The paper’s central negative documentation, developed in Section 4 as a sequence of five formal propositions, is that the standard retail Contract-for-Difference exhibits five structural incompatibilities or concerns under the adopted AAOIFI/OIC constraint framework: an interest-rate-linked and credit-plus- increment financing structure (Proposition 4.1); substantial *gharar* concerns arising from cash-settled contingent payoff without ownership or delivery, forced- liquidation risk, and asymmetric execution (Proposition 4.2); a zero-sum counterparty structure on the internalised fraction of notional under B-book execution (Proposition 4.3); non-fit with the classical nominate contract categories, including absence of *qabd* (Proposition 4.4); and a reciprocal-deferred-contingent obligation structure that engages the *bay’ al-kāli’ bi’l-kāli’* prohibition on the broad reading (Proposition 4.5). The “swap-free Islamic account” variant marketed by conventional brokers, analysed in Section 4.7, addresses at most the first of these concerns and only in part.

The paper’s central positive contribution, developed in Section 5.4, is a market-access architecture built on the classical *sharīkat al-’inān* joint-equity partnership contract, together with a proof that this architecture is formally consistent with the operational design constraints (D1)–(D5) adopted in this paper, subject to review by qualified Shariah supervisory boards (Proposition 5.6). The architecture provides the client with genuine economic exposure to a chosen deliverable underlying at unit leverage, with the broker acting as an agent-manager on the same side of the venture at the gross level rather than as an opposing counterparty (Remark 5.8). The absence of leverage is a structural property of this specific joint-ownership construction under the paper’s adopted constraints (Remark 5.9); the paper

does not claim that all Shariah-compliant financing anywhere is incapable of leverage.

The paper’s central simulation-based quantitative claim, established in Section 6 through Monte Carlo simulation under illustrative Bates parameterisations across four underlyings, four holding horizons, and four leverage regimes, is a quantification of the compliance cost of the Shariah leverage cap. Because the compliant architecture cannot be leveraged, its comparison with a leveraged CFD is not a like-for-like exposure comparison and does not support a superiority claim in the ordinary financial sense. What the simulation *does* establish is: (i) at unit exposure, the underlying-return component of the two architectures is closely matched, before architecture-specific fees, execution mechanics, and cash-flow definitions; (ii) the CFD’s fee structure scales with the leveraged notional whereas the *sharīkat* ujah does not, producing a crossover in absolute cost of carry at approximately $L \geq 5$ for holding horizons above one week; and (iii) at higher leverage regimes, the CFD introduces margin-call forced-liquidation risk that the unleveraged *sharīkat* structurally cannot exhibit. At $L = 10$ on the S&P 500 over a 63-day horizon, the leveraged CFD triggers margin-call events on 48% of paths; no maintenance-margin calls were observed in the 10,000 simulated *sharīkat* paths, and more fundamentally the *sharīkat* architecture contains no price-triggered maintenance-margin close-out mechanism. At $L = 30$ the CFD saturates the negative-balance protection floor on 75% of paths (Table 4). These figures describe the trade the Muslim participant faces when choosing between the architectures: reduced exposure in exchange for the elimination of the maintenance-margin close-out mechanism and formal consistency with the constraint set.

8.2. Contributions restated

The five contributions announced in Section 1.4 have been developed in this paper:

1. A formal characterisation of the standard CFD/FX architecture as a coupled stochastic system (Section 3).
2. Formal propositions documenting five structural incompatibilities or concerns under the adopted AAOIFI/OIC constraint framework (Section 4).
3. Three first-principles alternative architectures with explicit payoff dynamics, one of which (*sharīkat al-’inān*) is formally consistent with the operational design constraints adopted in this paper (Section 5).

4. A simulation-based quantitative comparison of all-in holding cost, tail risk, and margin-call frequency across architectures, framed as a compliance-cost analysis (Section 6).
5. An open-source Python reference implementation released under the MIT License (Section 6.8).

8.3. Directions for future research

The title of this paper, “Shariah Computational Financial Engineering I”, signals its role as the opening entry in a planned series. Subsequent papers in the series will address the following directions, each of which extends the framework of Definitions 4.1 and 5.1 into a new instrument class or methodological setting.

Paper II: Shariah-compliant portfolio optimisation under AAOIFI screening constraints. The classical Markowitz mean-variance framework requires modification to accommodate (i) sector exclusion constraints, (ii) financial-ratio screening thresholds under AAOIFI Standard No. 21, and (iii) *purification* of incidental impermissible income. The resulting optimisation problem is convex under natural formulations but requires purpose-built solver architecture; the theoretical analysis and empirical performance comparison against conventional benchmarks constitute the intended subject of the next paper.

Paper III: Şukūk pricing under stochastic profit-rate models. The equivalent of interest-rate modelling for the Shariah-compliant fixed income market. The analogue of the Heath–Jarrow–Morton framework under a profit-rate rather than interest-rate driver, calibrated against the growing *şukūk* market and applicable to profit-sharing investment accounts as well as issued *şukūk*.

Paper IV: Takāful risk modelling and Shariah-compliant actuarial reserving. Cooperative insurance under the *mudāraba* or *wakāla* structure requires actuarial treatment that departs from the standard mutual-of-strangers pooling model. Regulatory capital and reserving under a genuinely cooperative structure is an open technical question with substantial practical stakes.

Beyond the planned series, several natural extensions of the present paper’s approach suggest themselves:

- An empirical study of observant-Muslim retail-trader behaviour on Shariah-compliant versus conventional accounts, to test whether the leverage-cap feature of the *sharīkat* architecture improves realised outcomes as the theoretical analysis predicts.

- A deep-learning approach to Shariah-compliance classification for equity screening at scale, addressing the operational bottleneck faced by every Islamic asset manager.
- Formal analysis of systematic hedging strategies under the *wa'd*-based framework of Section 5.2, including cost-of-carry comparisons against conventional NDFs and Islamic FX Forward products.
- Extension of the market-access architectures to decentralised-finance settings, where the joint-ownership structure of Definition 5.4 maps naturally onto smart-contract-based tokenised custody.

8.4. Closing remark

The observant-Muslim participant base in global capital markets is large, rising, and structurally underserved by the current product landscape. Sophisticated tools for their genuine participation, under structures that respect the religious constraints that a substantial share of the world's population regards as binding, represent both an intellectual challenge and a real contribution to financial infrastructure. This paper takes one small step toward that broader research programme. Much more remains to be done.

Appendix A. Full mathematical derivations

This appendix collects the derivations underlying the propositions of Sections 3–5. All notation is as in the main text.

Appendix A.1. Conditional distribution of the log-price under Bates

Let $X_t := \ln(S_t/S_0)$ denote the log-price process under Assumption 3.1. Its characteristic function $\phi_X(u; t) := \mathbb{E}[\exp(iuX_t) \mid \mathcal{F}_0]$ admits the semi-closed decomposition

$$\phi_X(u; t) = \phi_X^{\text{Heston}}(u; t) \cdot \phi_X^{\text{jump}}(u; t), \quad (\text{A.1})$$

where the diffusive component follows the Heston (1993) closed form

$$\begin{aligned}
\phi_X^{\text{Heston}}(u; t) &= \exp\left(C(u; t) + D(u; t)v_0 + iu\mu t\right), \\
D(u; t) &= \frac{b_2 - d}{\sigma_v^2} \left(\frac{1 - e^{-dt}}{1 - ge^{-dt}} \right), \\
C(u; t) &= \frac{\kappa_v \theta}{\sigma_v^2} \left[(b_2 - d)t - 2 \ln \left(\frac{1 - ge^{-dt}}{1 - g} \right) \right], \\
b_2 &= \kappa_v - i\rho\sigma_v u, \\
d &= \sqrt{b_2^2 + \sigma_v^2(u^2 + iu)}, \\
g &= (b_2 - d)/(b_2 + d),
\end{aligned} \tag{A.2}$$

and the jump component is

$$\phi_X^{\text{jump}}(u; t) = \exp\left(\lambda t \left[e^{iu\mu_J - u^2\sigma_J^2/2} - 1 - iu\kappa \right]\right). \tag{A.3}$$

The conditional density of X_t is recovered by Fourier inversion,

$$f_{X_t}(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-iux} \phi_X(u; t) du, \tag{A.4}$$

which admits numerical evaluation via the Carr–Madan FFT scheme in settings where an analytic density is required (Bates, 1996; Björk, 2009). This Fourier machinery is included here for analytical completeness only. The Monte Carlo path simulation used in Section 6 integrates the discretised SDE of Assumption 3.1 directly and does not invoke the density or the FFT inversion.

Appendix A.2. Margin-call barrier hitting probability

We derive the closed-form margin-call probability quoted in Proposition 3.2. Suppose S_t follows the geometric Brownian motion special case of Assumption 3.1 (i.e. $\lambda = 0$, $v_t \equiv \sigma^2$). The log-price $X_t = \ln(S_t/S_{t_0})$ is a Brownian motion with drift:

$$X_t = \left(\mu - \frac{1}{2}\sigma^2\right)(t - t_0) + \sigma W_{t-t_0}, \quad t \geq t_0. \tag{A.5}$$

For a long CFD position, the margin-call barrier of (16) translates into the log-price barrier

$$b := \ln\left(1 - \frac{1 - \alpha}{L}\right) = -\frac{1 - \alpha}{L} + O(L^{-2}), \quad (\text{A.6})$$

so that $\tau_{\text{MC}} = \inf\{t > t_0 : X_t \leq b\}$.

The standard first-passage distribution for Brownian motion with drift (Shreve, 2004, Theorem 3.7.6) gives

$$\mathbb{P}(\tau_{\text{MC}} < h) = \Phi\left(\frac{b - \nu h}{\sigma\sqrt{h}}\right) + e^{2\nu b/\sigma^2} \Phi\left(\frac{b + \nu h}{\sigma\sqrt{h}}\right), \quad (\text{A.7})$$

where $\nu = \mu - \sigma^2/2$ and $h = t_1 - t_0$. Setting $\nu = 0$ (leading-order approximation on short horizons) and substituting the leading term of (A.6) yields

$$\mathbb{P}(\tau_{\text{MC}} < h) \approx 2\Phi\left(-\frac{1 - \alpha}{L\sigma\sqrt{h}}\right), \quad (\text{A.8})$$

which is the approximation quoted in Proposition 3.2. Writing $x := (1 - \alpha)/(L\sigma\sqrt{h})$, the leverage sensitivity follows from the standard Taylor expansion of the standard normal CDF near the origin,

$$\Phi(-x) = \frac{1}{2} - \frac{x}{\sqrt{2\pi}} + O(x^3), \quad x \rightarrow 0, \quad (\text{A.9})$$

which under $L \rightarrow \infty$ (hence $x \rightarrow 0$) gives

$$\mathbb{P}(\tau_{\text{MC}} < h) = 1 - \sqrt{\frac{2}{\pi}} \cdot \frac{1 - \alpha}{L\sigma\sqrt{h}} + O(L^{-3}) \rightarrow 1 \quad \text{as } L \rightarrow \infty. \quad (\text{A.10})$$

Note: the more familiar Mills-ratio asymptotic $\Phi(-x) \sim \phi(x)/x$ holds as $x \rightarrow \infty$; here we are in the opposite regime ($x \rightarrow 0$) and the expansion of (A.9) is the correct one. The limit $\mathbb{P} \rightarrow 1$ is intuitive: at infinite leverage the barrier coincides with the entry price and any downward Brownian excursion within horizon h triggers a margin call.

For the full Bates dynamics ($\lambda > 0$), (A.7) is *not* in general a strict bound in either direction: the log-jump-size distribution $\ln J_t \sim \mathcal{N}(\mu_J, \sigma_J^2)$ admits both $J_t > 1$ and $J_t < 1$, so a jump event can move the path either towards or away from a downward barrier. The direction and magnitude of

the effect depend on $(\lambda, \mu_J, \sigma_J)$ and the compensator adjustment $-\lambda\kappa$; the net contribution is quantified numerically in Section 6 (Figure 4).

Appendix A.3. Wa'd payoff derivation and cost-of-carry decomposition

Under Definition 5.2, at delivery time T the client and broker execute a spot FX transaction at the pre-agreed rate K . The client's cash flows in CCY2 are:

- At t_0 : pay ujah $u^{\text{wa'd}}$ (if front-loaded) or accrue it to T .
- At T : pay $N \cdot K$ units of CCY2, receive N units of CCY1.
- Immediately after T : sell N units of CCY1 at spot rate S_T , receive $N \cdot S_T$ units of CCY2.

Netting yields $\Pi_c^{\text{wa'd}}(t_0, T) = N(S_T - K) - u^{\text{wa'd}}$, which is equation (29).

Under the risk-neutral measure $\mathbb{P}^*$ implied by covered interest rate parity, the discounted forward rate satisfies $\mathbb{E}^*[S_T | \mathcal{F}_{t_0}] = F_0^{\text{CIP}}$, so

$$\mathbb{E}^*[\Pi_c^{\text{wa'd}}(t_0, T) | \mathcal{F}_{t_0}] = N(F_0^{\text{CIP}} - K) - u^{\text{wa'd}}. \quad (\text{A.11})$$

The economic cost of the wa'd relative to a conventional NDF struck at $K = F_0^{\text{CIP}}$ is thus exactly $u^{\text{wa'd}}$ in expected value. Under the physical measure $\mathbb{P}$, the comparison depends on the market's risk premium, which is not constrained by the wa'd mechanics but only by the joint calibration decision on K and $u^{\text{wa'd}}$ discussed in Remark 5.1.

Appendix A.4. Sharikat al-'inān payoff, extended derivation

We extend the proof of Proposition 5.5 to the general case of unequal profit and loss shares under the AAOIFI Standard No. 12 §3/1/5/2 constraint that loss shares equal capital contribution ratios.

Let π_c, π_b denote the profit-sharing ratios with $\pi_c + \pi_b = 1$, and let $\ell_c = K_c/(K_c + K_b)$, $\ell_b = K_b/(K_c + K_b)$ denote the loss-sharing ratios. The joint venture's terminal P&L is $\Delta := N \cdot (S_{t_1} - S_{t_0})$, where $N = (K_c + K_b)/S_{t_0}$. The client's share is

$$\Pi_c(\Delta) = \begin{cases} \pi_c \cdot \Delta - u_c^{\text{shr}} & \text{if } \Delta \geq 0, \\ \ell_c \cdot \Delta - u_c^{\text{shr}} & \text{if } \Delta < 0. \end{cases} \quad (\text{A.12})$$

Under the AAOIFI-compliant proportional case $\pi_c = \ell_c = K_c/(K_c + K_b)$, the two branches of (A.12) collapse to the single expression

$$\Pi_c = \frac{K_c}{K_c + K_b} \cdot N \cdot (S_{t_1} - S_{t_0}) - u_c^{\text{shr}} = K_c \cdot \left(\frac{S_{t_1}}{S_{t_0}} - 1 \right) - u_c^{\text{shr}}, \quad (\text{A.13})$$

recovering equation (35). In the general case with $\pi_c > \ell_c$ (higher profit share than loss share for one party, compensating that party for taking on disproportionate management responsibility or for non-capital contributions to the venture, per AAOIFI SS No. 12 §3/1/5), the payoff is a piecewise-linear call-like structure that may remain permissible under the applicable partnership rules, provided the loss-sharing constraint $\ell_c = K_c/(K_c + K_b)$ is satisfied (an AAOIFI SS No. 12 constraint that forbids losses being borne otherwise than in proportion to capital) and all other contractual requirements are met, subject to SSB review; the client’s expected profit is enhanced relative to the proportional case, with the excess taken from the broker’s profit share.

Appendix B. Fiqh sources catalogued

Table B.5 consolidates the primary fiqh sources cited across the paper. AAOIFI Shariah Standards are cited by number and, where relevant, by section reference; OIC Fiqh Academy resolutions by number and session; contemporary scholarly works by author and section reference within the cited text. This catalogue is intended as a navigation aid for scholarly review of the paper’s compliance claims.

Appendix C. Python code documentation

This appendix documents the module structure of the reference Python implementation released at <https://github.com/Muhammad-Taha-Asif/Shariah-CFE-1>. Each module is self-contained and can be imported independently.

src/bates.py. Implements Assumption 3.1. Exports:

- **BatesParams**: a frozen dataclass encoding the ten parameters of the Bates specification, with helper properties for the mean jump size and the Feller condition check.

Table B.5: Primary fiqh sources cited in this paper. AAOIFI SS denotes AAOIFI Shariah Standard; OIC Res. denotes International Islamic Fiqh Academy Resolution.

Source	Subject	Cited in
AAOIFI SS No. 1	Trading in Currencies; possession requirements, deferment prohibition, restrictions on forward/futures currency transactions and on binding bilateral currency promises (§2/9)	Prop. 4.1 (in support of the general <i>ribā</i> argument), Prop. 5.2 (constraint on Architecture A), Rem. 3.4
AAOIFI SS No. 8	<i>Murābaḥa</i> and credit-sale increments; prohibitions related to <i>bay' al-kāli' bi'l-kāli'</i>	Prop. 4.5, Table 1
AAOIFI SS No. 12 §3/1/5/2	<i>Sharīkah</i> ; proportional loss-sharing rule	Def. 5.4, App. Appendix A.4
AAOIFI SS No. 18	<i>Qabḍ</i> (constructive possession) requirement	Prop. 4.4, Table 1
AAOIFI SS No. 20	Sale of commodities in organised markets	Section 2.2
AAOIFI SS No. 21	Financial-ratio screening for Shariah-compliant equities	Section 5.4, Section 8.3
AAOIFI SS No. 27	Indices	Section 2.2
AAOIFI SS No. 30 §5	Organised <i>tawarruq</i>	Prop. 5.4
AAOIFI SS No. 31 §5/1	Controls on <i>gharar</i> in financial transactions	Prop. 4.2, Table 1
OIC Res. 63 (1/7), 1992	Financial markets; definition of <i>maysir</i>	Prop. 4.3, Table 1
OIC Res. 179 (19/5), 2009	Organised <i>tawarruq</i> ; impermissibility ruling	Prop. 5.4, Rem. 5.3
OIC Res. 238 (24/2), 2019	Mutual binding commitments in future currency exchange	Rem. 5.1, Prop. 5.2
AAOIFI SS No. 49	Unilateral and Bilateral Promise (<i>al-wa'd</i> and <i>al-muwā'adah</i>)	Rem. 5.1
AAOIFI Governance Standard No. 1	Shariah Supervisory Board: appointment, composition, and report	Rem. 5.7
Ḥadīth of Ibn Umar	Prophetic prohibition against <i>bay' al-kāli' bi'l-kāli'</i>	Prop. 4.5
Qur'ān 2:275	Prohibition of <i>ribā</i>	Section 2.1
Qur'ān 5:90	Prohibition of <i>maysir</i>	Section 2.1
Usmani (2002), ch. 3	Contemporary treatment of <i>ribā</i>	Prop. 4.1, Rem. 4.9
Kamali (2000), ch. 5	Contemporary treatment of <i>gharar</i> thresholds	Prop. 4.2
El-Gamal (2006)	Islamic finance law, economics, and practice	Rem. 5.1, Rem. 4.9

- `simulate_paths(params, n_paths, horizon_years, n_steps, seed)`: vectorised Monte Carlo simulator using Euler–Maruyama on the log-price with full-truncation of the variance process to preserve positivity, following Andersen (2008). Returns paired arrays of price and variance paths.
- `preset(name)`: returns the illustrative calibration presets of Table 3 for “EURUSD”, “USDJPY”, “AUDUSD”, “SPX”.

src/contracts.py. Implements the CFD contract mechanics of Section 3 and the *sharīkat* architecture of Section 5.4. Exports:

- `CFDParams, SharikaParams`: dataclasses encoding contract parameters.
- `cfid_payoff_with_margin_call`: computes CFD gross P&L, accumulated Tom-Next financing, and net client P&L accounting for margin-call forced liquidation and negative-balance protection.
- `margin_call_time`: returns the first-hitting index of the barrier of equation (16) for each simulated path.
- `sharika_payoff`: computes the client’s net payoff under Proposition 5.5.

src/metrics.py. Statistical machinery for Section 6.2. Exports `moments`, `var_cvar`, `sharpe`, `sortino`, `calmar`, `bootstrap_ci`, `ks_test`, and `summarise`. All operate on one-dimensional P&L arrays with the sign convention that positive values are gains.

src/simulate.py. Master grid runner for Section 6. Iterates over four underlyings, four horizons, four leverages, and runs `run_cell()` on each combination, persisting the results to `tables/results_grid.csv`. All random seeds are derived from a session-wide `SEED_BASE = 20260928` for exact reproducibility.

src/figures.py. Publication-quality figure generation for Section 6.5. Produces `fig1_cost_of_carry.pdf` through `fig4_tail_decomposition.pdf` with typography configured to match the paper’s Latin Modern Roman body text.

src/ml_analysis.py. XGBoost margin-call classifier of Section 6.4. Trains the classifier on early-path features and generates *fig5_feature_importance.pdf*.

Dependencies are declared in *requirements.txt* at the repository root. The full simulation grid runs in approximately two minutes on commodity laptop hardware.

Appendix D. Data sources and calibration protocol

The Monte Carlo results in Section 6 are generated under the illustrative Bates parameter presets of Table 3. These presets are consistent in sign and order of magnitude with published FX and equity implied-volatility studies (Andersen, 2008; Bates, 1996) and with contemporaneous option-market observations for the four underlyings, but are not the result of a proprietary calibration.

For a production implementation of the analysis, the recommended data pipeline is as follows.

Tick-level FX data.. Free public tick data for the three FX pairs is available from Dukascopy Bank SA (Geneva) via their historical data feed. Data covering the period 2015–2025 is required to span the low-volatility regime (2015–2018), the moderate-volatility period (2019–2022), and the elevated regime (2022–2025). Data acquisition should respect Dukascopy’s terms of service, which permit academic research use but not onward redistribution; a download script for the recommended snapshots is included in the repository (*data/* directory) but no raw data are checked in.

Equity and index data.. Historical minute-bar data for the S&P 500 index and its ETF proxies (SPY, VOO) is available from the CBOE DataShop or, for daily data, Yahoo! Finance via the *yfinance* Python package. Bid-ask spread data at the tick level requires a paid data vendor (TAQ, Bloomberg B-Pipe).

Interbank overnight rates.. Data for SOFR, ESTR, SONIA, TONA, and the RBA cash rate is available via the respective central bank public data portals at daily frequency; intraday granularity, where needed for cross-checking with Tom-Next swap rates, is available from the OIS market data providers.

Data cleaning protocol.. A production calibration would apply the following cleaning steps in order: (i) exclude weekend and public-holiday timestamps; (ii) exclude tick-level records with zero volume, zero spread, or a bid–ask cross; (iii) apply a *jump-preserving* tick-cleaning procedure, such as the Brownlees and Gallo (2006) algorithm or a comparable rule that removes microstructure noise (mistaken ticks, feed errors, quotes with implausible spreads) without deleting genuine economic jumps, a blanket $\pm 5\sigma$ filter is expressly not applied at this stage, since the Bates specification aims to *estimate* jump-intensity and jump-size parameters and a naive tail-cut would bias λ , μ_J , and σ_J downward; (iv) resample from tick to five-minute bars to reduce microstructure noise while preserving jump signatures; (v) construct daily aggregate return, realised variance, and realised skewness series for calibration input; the distinction between bad-tick outliers and genuine jumps in the resulting series is then handled inside the Bates calibration itself rather than by pre-filtering.

Parameter calibration.. The Bates parameters are recoverable from either (a) time-series estimation using particle filtering or MCMC on the coupled SDE system, or (b) cross-sectional calibration against traded option-market implied volatility surfaces via Fourier pricing on the characteristic function of Appendix A.1. The latter approach is preferred where an option market exists for the underlying; for underlyings without a liquid option market, time-series calibration is the default. Regular recalibration (weekly or on material market events) is recommended.

Full implementation of the calibration pipeline is deferred to Paper II of this series (Section 8.3) and is not required for the architectural comparison that is the subject of the present paper.

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